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SCHOOL: School of Internship Training

DEPARTMENT: Digital Forensics

ACADEMY: ICDFA

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COURSE TITLE: Computer Forensics Case Study I

LAB TITLE: Rhino Hunting: USB, Steganography and
Network Evidence

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EXECUTIVE SUMMARY

This digital forensic examination was conducted as part of the ICDFA training case study identified as CIP-B104-CS3-C11/26/DFIT/17297. The examination focused on the identification, recovery, validation, correlation, and classification of digital evidence associated with a historical case involving multiple rhinoceros images. The investigation was designed to determine whether the available digital evidence contained sufficient information to establish the presence, transfer, storage, recovery, and potential possession of multiple unique rhinoceros images across different evidence sources.

The examination considered four principal evidence items contained within the supplied evidence package: RHINOUSB.dd, rhino.log, rhino2.log, and rhino3.log. The evidence package itself was preserved and examined in a controlled forensic working environment. Cryptographic SHA-256 hashes were calculated for the principal evidence files to establish and document their integrity before analysis. The forensic methodology maintained a clear distinction between original evidence and working/recovered material, with the original evidence treated as read-only and all investigative processing performed on copies or exported derivatives.

The USB disk image, RHINOUSB.dd, was examined to establish its filesystem characteristics and identify both allocated and recoverable content. The disk image was identified as a FAT16 volume with a 512-byte sector size, 8 sectors per cluster, two FATs, 512 root-directory entries, and 248 sectors per FAT. Examination of the root directory identified two allocated text files, GUMBO1.TXT and GUMBO2.TXT. These files contained recipe-related textual material and repeated textual strings. While these artifacts were documented as part of the filesystem

examination, their presence alone was not treated as sufficient evidence of rhinoceros-image possession or attribution.

File-carving analysis of the USB image identified multiple JPEG and GIF signatures. Eight image objects were successfully carved and validated as readable image files. Subsequent visual examination established that several of these recovered objects were unrelated wildlife images, including crocodile and lizard imagery. Importantly, two of the carved JPEG objects were identified as depicting rhinoceroses. One was a 169×228 JPEG with a SHA-256 hash of 568457ed594c4e98dc503e3e244b1e316376aafb0d1c283d95601d13762a748c, while another was a 1024×685 JPEG with a SHA-256 hash of b4f6bbb8d846d6505c9b5b67f115656fdf980a484aa441078a838a3ffc2ac705. These files constitute significant recovered evidence from the USB image.

The carving examination also demonstrated the importance of cryptographic deduplication. At least one recovered JPEG occurred more than once within the disk image, with identical SHA-256 values establishing that the duplicate occurrences represented the same digital file content rather than two independent images. Consequently, the duplicate was not counted as an additional unique rhinoceros image. This approach prevents inflation of the final image count and provides a defensible basis for distinguishing unique digital objects from repeated instances.

Network forensic analysis of rhino.log identified FTP activity involving the client system and an FTP server. Examination of the FTP control and data communications identified multiple file-transfer operations associated with rhinoceros-image filenames and an encrypted ZIP archive. The first relevant FTP transfer resulted in recovery of rhino1.jpg. The recovered JPEG was

validated as a legitimate image and visually confirmed to depict a rhinoceros. Its SHA-256 hash was recorded as 0f30a8777d19a4fe4b2fafd5154b6c4dfafb673202db29971e876e978da4e05.

A subsequent FTP session contained transfers of rhino3.jpg. Two data-stream instances associated with the same filename were identified, and both produced identical file content with a SHA-256 hash of a85aea8e47d9f5425456f3bafb9c67201c8c2b2b2cec80154c83102e802240c5b. The recovered image was visually confirmed as a rhinoceros image. Because the two transfers produced identical cryptographic hashes, they were treated as duplicate transfers of the same digital object rather than two unique rhinoceros images. This distinction is particularly important when calculating the number of unique rhinoceros images represented in the evidence.

The FTP examination also identified an encrypted archive named contraband.zip. The archive was recovered directly from network traffic and validated as a ZIP-format file. Structural examination confirmed that the archive contained a member named rhino2.jpg and that the archive was encrypted. The archive's member information showed an uncompressed size of 230,665 bytes. Notably, this size corresponds exactly with the size of one of the recovered JPEG objects carved from the USB image. This provides a strong correlation between the USB-carved image and the archived object; however, definitive identification as the same digital object requires extraction of rhino2.jpg followed by cryptographic hash comparison. The controlled dictionary-recovery procedure described in the supplied Module 3 material provides the appropriate forensic workflow for recovering the archive password and extracting the encrypted member. The recovered password itself should remain within the protected forensic working records and should not be unnecessarily reproduced in the main report.

The network examination of rhino2.log identified HTTP activity involving a web server and a client system. The HTTP evidence included requests for rhino4.jpg and rhino5.gif. Reassembly of the relevant HTTP response streams resulted in recovery of a JPEG corresponding to rhino4.jpg and a GIF corresponding to rhino5.gif. The recovered rhino4.jpg object was a 1531 × 2286 JPEG with SHA-256 hash e1c01da391669adf3559d2d87bf2157b37e27a936a1ae66c77b61e6b5b7bc4a6 and was visually confirmed to depict a rhinoceros. The rhino5.gif object was a 400 × 275 GIF with SHA-256 hash f36ecc12967d5622cdc70e73d2468fab28a65db944d3157256a7212afa05b056 and was likewise visually confirmed as depicting a rhinoceros. Smaller GIF objects appearing in the same HTTP response streams were treated as unrelated supporting web content rather than automatically being classified as evidence of rhinoceros imagery.

The rhino3.log network capture contains additional web activity, including searches and HTTP communications associated with executable files. A likely rhino.exe transfer was identified within the capture. Because executable files can present significant security and evidentiary risks, the appropriate forensic approach is static examination rather than execution on an ordinary workstation. The executable should be exported from the relevant network stream, hashed, identified, and examined using approved static-analysis tools. No conclusion concerning execution, malicious functionality, or user attribution should be drawn solely from its presence in network traffic without additional supporting evidence.

The examination also considered the requirement to investigate possible steganographic content. Detection of anomalous characteristics in an image file, unusual metadata, statistical irregularities, or suspicious padding can constitute an investigative lead, but such indicators

alone do not establish that information is hidden within an image. A defensible steganographic conclusion requires preservation of the original carrier, application of the approved detection method, controlled recovery where justified, validation of any extracted object, and cryptographic comparison against other evidence sources. Therefore, any final statement concerning hidden content should be based on the results of the assigned steganography workflow rather than on visual appearance, file padding, repeated strings, or signature anomalies alone.

Overall, the examination demonstrates that the evidence package contains rhinoceros-related digital artifacts distributed across multiple independent sources, including a FAT16 USB image, FTP communications, an encrypted ZIP archive, and HTTP traffic. Several of the recovered images are demonstrably unique based on their cryptographic hashes, while other occurrences represent repeated transfers or duplicate file instances. The evidence therefore supports a finding that multiple rhinoceros images were present within or transferred through the examined digital environment. The final numerical count of unique rhinoceros images should be established only after completion of the outstanding steganography examination, encrypted-archive extraction and hash comparison, executable/static-analysis documentation, and complete cross-source reduplications.

```

(kali@aivtic-students-kali)-[~/LABS]
$ sudo apt update

sudo apt install -y \
sleuthkit \
testdisk \
foremost \
scalpel \
wireshark \
tshark \
binwalk \
exiftool \
p7zip-full \
unzip \
zipinfo \
john \
fcrackzip \
strings \
file \
hashdeep \
imagemagick

[sudo] password for kali:
Sorry, try again.
[sudo] password for kali:
Sorry, try again.
[sudo] password for kali:
Get:1 http://kali.download/kali kali-last-snapshot InRelease [34.0 kB]
Err:1 http://kali.download/kali kali-last-snapshot InRelease
  The following signatures couldn't be verified because the public key is not available: NO_PUBKEY ED65462EC8D5E4C5
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
2382 packages can be upgraded. Run 'apt list --upgradable' to see them.

```

```

(kali@aivtic-students-kali)-[~/LABS]
$ sudo apt install imagemagick
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
aha appstream apt-config-icons apt-config-icons-hidpi apt-config-icons-large apt-config-icons-large-hidpi baloo-kf5 bluez-obexd breeze breeze-cursor-theme
breeze-gtk-theme breeze-icon-theme bup bup-doc catdoc cryfs dconf-cli docbook-xml docbook-xsl drkonqi espeak-ng-data evince-common
evolution-data-server-common firebird3.0-common firebird3.0-common-doc fonts-dejavu fonts-hack fonts-wine frameworkintegration geoclue-2.0
gir1.2-accountsservice-1.0 gir1.2-atspi-2.0 gir1.2-cloudproviders-0.3.0 gir1.2-freedesktop-dev gir1.2-gck-1 gir1.2-gcr-3 gir1.2-gdm-1.0 gir1.2-geoclue-2.0
gir1.2-glib-2.0-dev gir1.2-gmenu-3.0 gir1.2-gnomeautoar-0.1 gir1.2-gnomebg-4.0 gir1.2-gnomebluetooth-3.0 gir1.2-gtksource-4 gir1.2-gweather-4.0
gir1.2-ibus-1.0 gir1.2-javascriptcoregtk-4.1 gir1.2-javascriptcoregtk-6.0 gir1.2-json-1.0 gir1.2-mutter-13 gir1.2-nma4-1.0 gir1.2-packagekitglib-1.0
gir1.2-peas-1.0 gir1.2-polkit-1.0 gir1.2-rsvg-2.0 gir1.2-soup-3.0 gir1.2-totemlparsers-1.0 gir1.2-upowerglib-1.0 gir1.2-wnck-3.0 gkbd-capplet
gnome-backgrounds gnome-bluetooth-3-common gnome-session-bin gnome-session-common i965-va-driver i965-va-driver:i386 ibus ibus-data ibus-gtk ibus-gtk3
ibus-gtk4 im-config intel-media-va-driver intel-media-va-driver:i386 kaccounts-providers kamera kde-cli-tools kde-cli-tools-data kde-config-gtk-style
kde-config-sddm kde-config-updates kde-style-breeze kde-style-oxygen-qt5 kdeconnect kdegraphics-thumbnailers kdoctools5 kgamma5 khotkeys khotkeys-data
kimageformat-plugins kio-extras-data kirigami-addons-data kmenuedit kpeople-vcards kscreen kup-backup kwin-style-breeze kwin-wayland kwrited liba52-0.7.4
libaa1 libaa1:i386 libabs120220623:i386 libaccounts-glib0 libaccounts-qt5-1 libao-common libao4 libao3:i386 libappimagel1.0abi1 libappstreamqt5-3
libaribb24-0 libasound2:i386 libass9 libasynctns0:i386 libatk-adaptor libatk-bridge2.0-0:i386 libatk-bridge2.0-dev libatk1.0-0:i386 libatk1.0-dev
libatomic1:i386 libatspi2.0-0:i386 libatspi2.0-dev libavahi-client3:i386 libavahi-common-data:i386 libavahi-common3:i386 libavc1394-0 libavc1394-0:i386
libavif16:i386 libavif16-dev libavutil58 libavutil58:i386 libbcb729-0 libblkid-dev libblkid1:i386 libboost-chrono1.83.0 libboost-filesystem1.83.0
libboost-program-options1.83.0 libboost-thread1.83.0 libbrotli1.0.8 libbrotli-dev libbrotli1:i386 libbs2b0 libbsd0:i386 libbson-1.0-0 libbz2-1.0:i386
libbz2-dev libcac0 libcac0:i386 libcairo-gobject2:i386 libcairo2:i386 libcairo2-dev libcamel-1.2-64 libcap2:i386 libccdb2 libcdio-cdda2
libcdio-paranoia2 libcdio19 libcdparanoia0 libcdparanoia0:i386 libchromaprint1 libcurl4:i386 libcloudproviders-dev libcloudproviders0:i386 libcodecs2-1.2
libcodecs2-1.2:i386 libcolorcorrect5 libcolor2:i386 libcom-err2:i386 libcue2 libcupst2:i386 libcurl4:i386 libdatrie-dev libdatrie1:i386 libdav1d7:i386
libdb5-3:i386 libdbus-1-3:i386 libdc1394-25 libdca0 libde265-0:i386 libdehuginfd-common libdecor-0-0 libdecor-0-0:i386 libdecor-0-plugin-1-gtk:i386

```

```

(kali@aivtic-students-kali)-[~/LABS]
$ export CASE="CIP-B104-CS3-C11/26/DFIT/17297"

mkdir -p "$CASE"/{original,working,recovered,traffic-export,reports,screenshots,hashes,logs,appendix,tools}

(kali@aivtic-students-kali)-[~/LABS]
$ ls
CIP-B104-CS3-C11

(kali@aivtic-students-kali)-[~/LABS]
$ wget -q https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Illegal_Possession_Images/USB_image/DFRWS2005-RODEO.zip

(kali@aivtic-students-kali)-[~/LABS]
$ sha256sum DFRWS2005-RODEO.zip
md5sum DFRWS2005-RODEO.zip
e83459a80baed6a3c73551d413ab59fe8dac54c879ed5408001cfed171a0aaa DFRWS2005-RODEO.zip
6a80946a0fe694c12683a91019d6d2ef DFRWS2005-RODEO.zip

(kali@aivtic-students-kali)-[~/LABS]
$

```

```

(kali@aivtic-students-kali)-[~/LABS]
$ unzip DFRWS2005-RODEO.zip -d "$CASE/original"
Archive:  DFRWS2005-RODEO.zip
  inflating: CIP-B104-CS3-C11/26/DFIT/17297/original/RHINOUSB.dd
  inflating: CIP-B104-CS3-C11/26/DFIT/17297/original/rhino.log
  inflating: CIP-B104-CS3-C11/26/DFIT/17297/original/rhino2.log
  inflating: CIP-B104-CS3-C11/26/DFIT/17297/original/rhino3.log

(kali@aivtic-students-kali)-[~/LABS]
$ ls -l
total 3536
drwxr-xr-x 3 kali kali    4096 Sep 12 01:27 CIP-B104-CS3-C11
-rw-r--r-- 1 kali kali 3614418 Sep 12 01:28 DFRWS2005-RODEO.zip

(kali@aivtic-students-kali)-[~/LABS]
$

```

```

(kali@aivtic-students-kali)-[~/LABS]
$ unzip DFRWS2005-RODEO.zip
Archive:  DFRWS2005-RODEO.zip
  inflating: RHINOUSB.dd
  inflating: rhino.log
  inflating: rhino2.log
  inflating: rhino3.log

(kali@aivtic-students-kali)-[~/LABS]
$ ls -l
total 260592
drwxr-xr-x 3 kali kali    4096 Sep 12 01:27 CIP-B104-CS3-C11
-rw-r--r-- 1 kali kali    3614418 Sep 12 01:28 DFRWS2005-RODEO.zip
-rw-r--r-- 1 kali kali 259506176 Apr 30 2004 RHINOUSB.dd
-rw-r--r-- 1 kali kali   3187907 Apr 26 2004 rhino.log
-rw-r--r-- 1 kali kali   2926004 Apr 28 2004 rhino2.log
-rw-r--r-- 1 kali kali   2260994 Apr 28 2004 rhino3.log

(kali@aivtic-students-kali)-[~/LABS]
$

```

```

(kali@aivtic-students-kali)-[~/LABS]
$ sha256sum "$CASE/original/*"
md5sum "$CASE/original/*"
ce550424200a997c61b413941c8ef4df9619a2f96579674952294a176a32be65 CIP-B104-CS3-C11/26/DFIT/17297/original/RHINOUSB.dd
64e6d55b76660eb3aaa41572c1d04e4452510a343bbf42e844424827dedfddb2 CIP-B104-CS3-C11/26/DFIT/17297/original/rhino.log
41939d5de0556b70279056572dee44b6fd84cd05b0788cfd0b6f52f37b161dde CIP-B104-CS3-C11/26/DFIT/17297/original/rhino2.log
7b0304f5e88a30c305a99b5a1e2977bced5b9c94da6f458f887bb39966bbfc46 CIP-B104-CS3-C11/26/DFIT/17297/original/rhino3.log
80348c58eec4c328ef1f7709adc56a54 CIP-B104-CS3-C11/26/DFIT/17297/original/RHINOUSB.dd
c0d0093eb166acd7b73fa5225ae3f30 CIP-B104-CS3-C11/26/DFIT/17297/original/rhino.log
cd21eaf4acf50f71ffff857d7968341 CIP-B104-CS3-C11/26/DFIT/17297/original/rhino2.log
7e29f9d67346df25faaf18efcd95fc30 CIP-B104-CS3-C11/26/DFIT/17297/original/rhino3.log

(kali@aivtic-students-kali)-[~/LABS]
$

```

Evidence SHA-256

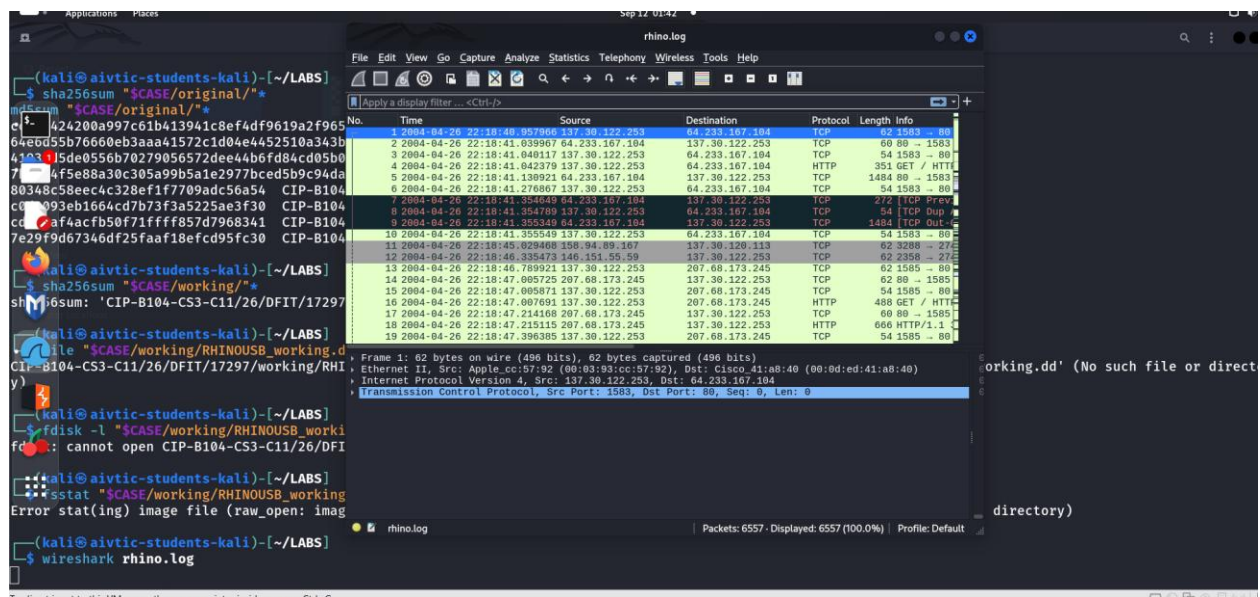
RHINOUSB.dd ce550424200a997c61b413941c8ef4df9619a2f96579674952294a176a32be65

rhino.log 64e6d55b76660eb3aaa41572c1d04e4452510a343bbf42e844424827dedfddb2

rhino2.log 41939d5de0556b70279056572dee44b6fd84cd05b0788cfd0b6f52f37b161dde

rhino3.log 7b0304f5e88a30c305a99b5a1e2977bced5b9c94da6f458f887bb39966bbfc46

Evidence ID	Description	Source	Method	Hash	Purpose
E01	RHINOUSB.dd	Original package	Supplied image	SHA-256	USB examination
E02	rhino.log	Original package	Supplied PCAP	SHA-256	FTP/network
E03	rhino2.log	Original package	Supplied PCAP	SHA-256	HTTP/email
E04	rhino3.log	Original package	Supplied PCAP	SHA-256	Executable/network
E05	carved images	E01	PhotoRec	SHA-256	Deleted recovery
E06	FTP images	E02	TCP reconstruction	SHA-256	Network evidence
E07	ZIP	E02	TCP reconstruction	SHA-256	Archive
E08	HTTP images	E03	HTTP export	SHA-256	Network evidence



rhino.log

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
2	2004-04-26 22:18:41.039967	64.233.107.104	137.30.122.253	TCP	60	62 1655 -> 21 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
3	2004-04-26 22:18:41.040117	137.30.122.253	64.233.107.104	TCP	60	62 21 -> 1655 [SYN, ACK] Seq=0 Ack=1 Min=49640 Len=0 MSS=1460 SACK_PERM
4	2004-04-26 22:18:41.042379	137.30.122.253	64.233.107.104	TCP	54	54 1655 -> 21 [ACK] Seq=1 Ack=1 Win=64240 Len=0
5	2004-04-26 22:18:41.130921	64.233.107.104	137.30.122.253	FTP	82	82 Response: 220 cook FTP server ready.
6	2004-04-26 22:18:41.276867	137.30.122.253	64.233.107.104	TCP	54	54 1655 -> 21 [ACK] Seq=1 Ack=29 Win=64212 Len=0
7	2004-04-26 22:18:41.354649	64.233.107.104	137.30.122.253	FTP	66	66 Request: USER gnome
8	2004-04-26 22:18:41.354768	137.30.122.253	64.233.107.104	TCP	60	60 21 -> 1655 [ACK] Seq=29 Ack=13 Win=49640 Len=0
9	2004-04-26 22:18:41.355349	64.233.107.104	137.30.122.253	FTP	88	88 Response: 331 Password required for gnome.
10	2004-04-26 22:18:41.355549	137.30.122.253	64.233.107.104	TCP	54	54 1655 -> 21 [ACK] Seq=13 Ack=63 Win=64178 Len=0
11	2004-04-26 22:18:45.029468	158.194.89.167	137.30.122.253	TCP	60	60 Request: PASS gnome123
12	2004-04-26 22:18:46.335473	146.151.95.59	137.30.122.253	TCP	60	60 21 -> 1655 [ACK] Seq=63 Ack=20 Win=49640 Len=0
13	2004-04-26 22:18:46.789921	137.30.122.253	207.68.173.24	FTP	81	81 Response: 230 User gnome logged in.
14	2004-04-26 22:18:47.005725	207.68.173.245	137.30.122.253	TCP	54	54 1655 -> 21 [ACK] Seq=28 Ack=90 Win=64151 Len=0
15	2004-04-26 22:18:47.005971	137.30.122.253	207.68.173.24	FTP	62	62 Request: TYPE I
16	2004-04-26 22:18:47.007691	137.30.122.253	207.68.173.24	FTP	74	74 Response: 200 Type set to I.
17	2004-04-26 22:18:47.214108	207.68.173.245	137.30.122.253	TCP	54	54 1655 -> 21 [ACK] Seq=36 Ack=110 Win=64131 Len=0
18	2004-04-26 22:18:47.215115	207.68.173.245	137.30.122.253	TCP	62	62 RTP Retransmission 4075 Seq=2245 [SYN] Seq=0 Win=16384 Len=0 MSS=1460 SACK_PERM
19	2004-04-26 22:18:47.488038	137.30.122.253	207.68.173.24	FTP	84	84 Response: 200 PORT command successful.
20	2004-04-26 22:18:47.505929	207.68.173.245	137.30.122.253	FTP	71	71 Request: STOR rhino1.jpg
21	2004-04-26 22:18:47.505929	207.68.173.245	137.30.122.253	TCP	62	62 20 -> 1657 [SYN] Seq=0 Win=49640 Len=0 MSS=1460 SACK_PERM
22	2004-04-26 22:18:47.505126	137.30.122.253	207.68.173.24	TCP	62	62 20 -> 1657 [SYN] Seq=0 Win=49640 Len=0 MSS=1460 SACK_PERM

Frame 1: 62 bytes on wire (496 bits), 62 bytes captured (496 bits) on Ethernet II, Src: Apple_cc:57:92 (08:03:93:cc:57:92), Dst: Cisco_41: Internet Protocol Version 4, Src: 137.30.122.253, Dst: 64.233.107.104, Transmission Control Protocol, Src Port: 1653, Dst Port: 80, Seq: 0

TCP, PERM
SS=1460
TCP segment of a reassembled PDU
[H, ACK] Seq=2861 Ack=298 Win=2920 Len=218 [TCP segment of a reasse
:1 Win=2810 Len=0
98 Win=2920 Len=1438
SACK_PERM
SACK_PERM
ACK_PERM
SS=1460 SACK_PERM
n=0
n=0

Close

rhino.log

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1526	2004-04-26 22:21:39.969911	137.30.122.253	137.30.120.40	TCP	62	62 1655 -> 21 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
1527	2004-04-26 22:21:39.970198	137.30.120.40	137.30.122.253	TCP	62	62 21 -> 1655 [SYN, ACK] Seq=0 Ack=1 Min=49640 Len=0 MSS=1460 SACK_PERM
1528	2004-04-26 22:21:39.970250	137.30.120.40	137.30.120.40	TCP	54	54 1655 -> 21 [ACK] Seq=1 Ack=1 Win=64240 Len=0
1529	2004-04-26 22:21:39.998180	137.30.120.40	137.30.122.253	FTP	82	82 Response: 220 cook FTP server ready.
1530	2004-04-26 22:21:40.147439	137.30.122.253	137.30.120.40	TCP	54	54 1655 -> 21 [ACK] Seq=1 Ack=29 Win=64212 Len=0
1531	2004-04-26 22:21:43.596019	137.30.122.253	137.30.120.40	TCP	62	62 4075 -> 2245 [SYN] Seq=0 Win=16384 Len=0 MSS=1460 SACK_PERM
1532	2004-04-26 22:21:43.596613	137.30.122.253	137.30.120.40	FTP	66	66 Request: USER gnome
1533	2004-04-26 22:21:43.598613	137.30.120.40	137.30.122.253	TCP	60	60 21 -> 1655 [ACK] Seq=29 Ack=13 Win=49640 Len=0
1534	2004-04-26 22:21:43.602936	137.30.120.40	137.30.122.253	FTP	88	88 Response: 331 Password required for gnome.
1535	2004-04-26 22:21:43.758957	137.30.122.253	137.30.120.40	TCP	54	54 1655 -> 21 [ACK] Seq=13 Ack=63 Win=64178 Len=0
1536	2004-04-26 22:21:45.025720	137.30.122.253	137.30.120.40	FTP	60	60 Request: PASS gnome123
1537	2004-04-26 22:21:45.025918	137.30.120.40	137.30.122.253	TCP	60	60 21 -> 1655 [ACK] Seq=63 Ack=20 Win=49640 Len=0
1538	2004-04-26 22:21:45.760912	137.30.120.40	137.30.122.253	FTP	81	81 Response: 230 User gnome logged in.
1539	2004-04-26 22:21:45.865674	137.30.122.253	137.30.120.40	TCP	54	54 1655 -> 21 [ACK] Seq=28 Ack=90 Win=64151 Len=0
1540	2004-04-26 22:21:46.569519	137.30.122.253	137.30.120.40	FTP	62	62 Request: TYPE I
1541	2004-04-26 22:21:46.569784	137.30.120.40	137.30.122.253	FTP	74	74 Response: 200 Type set to I.
1542	2004-04-26 22:21:46.668224	137.30.122.253	137.30.120.40	TCP	54	54 1655 -> 21 [ACK] Seq=36 Ack=110 Win=64131 Len=0
1543	2004-04-26 22:21:46.680481	120.130.102.117	137.30.120.40	TCP	62	62 RTP Retransmission 4075 Seq=2245 [SYN] Seq=0 Win=16384 Len=0 MSS=1460 SACK_PERM
1544	2004-04-26 22:21:49.952000	137.30.120.40	137.30.122.253	FTP	81	81 Request: PORT 137.30.122.253,6,123
1545	2004-04-26 22:21:49.953485	137.30.120.40	137.30.122.253	TCP	84	84 Response: 200 PORT command successful.
1546	2004-04-26 22:21:49.954047	137.30.122.253	137.30.120.40	FTP	71	71 Request: STOR rhino1.jpg
1547	2004-04-26 22:21:49.954833	137.30.120.40	137.30.122.253	TCP	62	62 20 -> 1657 [SYN] Seq=0 Win=49640 Len=0 MSS=1460 SACK_PERM

Frame 1536: 60 bytes on wire (552 bits), 60 bytes captured (552 bits) on Ethernet II, Src: Apple_cc:57:92 (08:03:93:cc:57:92), Dst: Oracle_fc:13:96 (08:00:20:fe:13:96) Internet Protocol Version 4, Src: 137.30.122.253, Dst: 137.30.120.40, Transmission Control Protocol, Src Port: 1655, Dst Port: 21, Seq: 13, Ack: 63, Len: 15 File Transfer Protocol (FTP) - PASS gnome123\r\n Request command: PASS Request arg: gnome123 [Current working directory:]

rhino.log

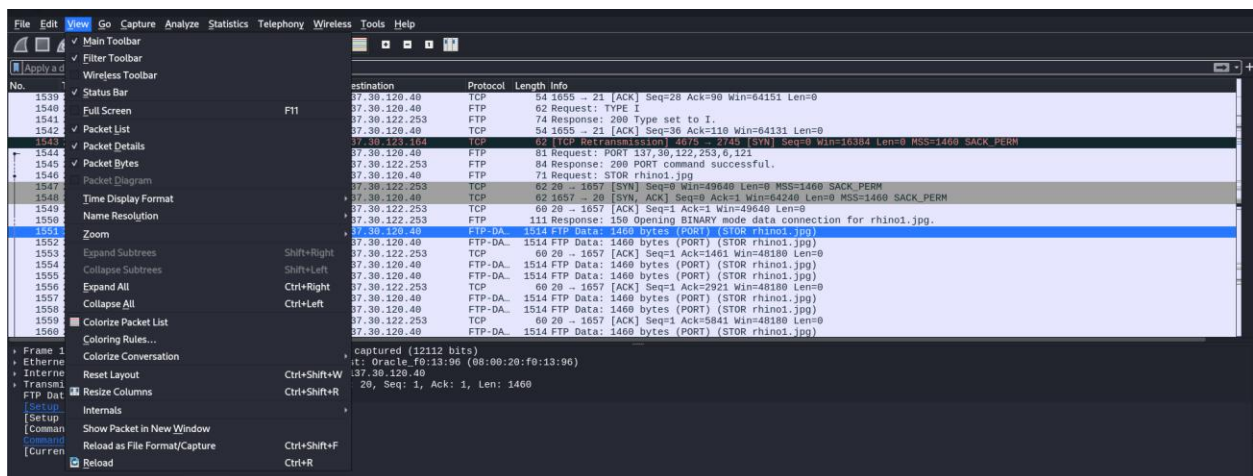
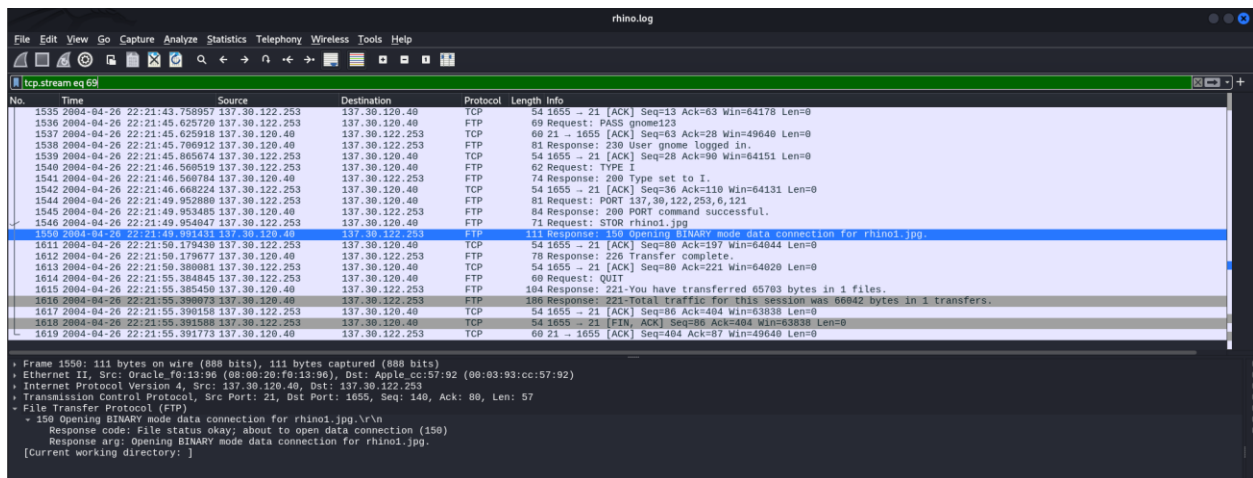
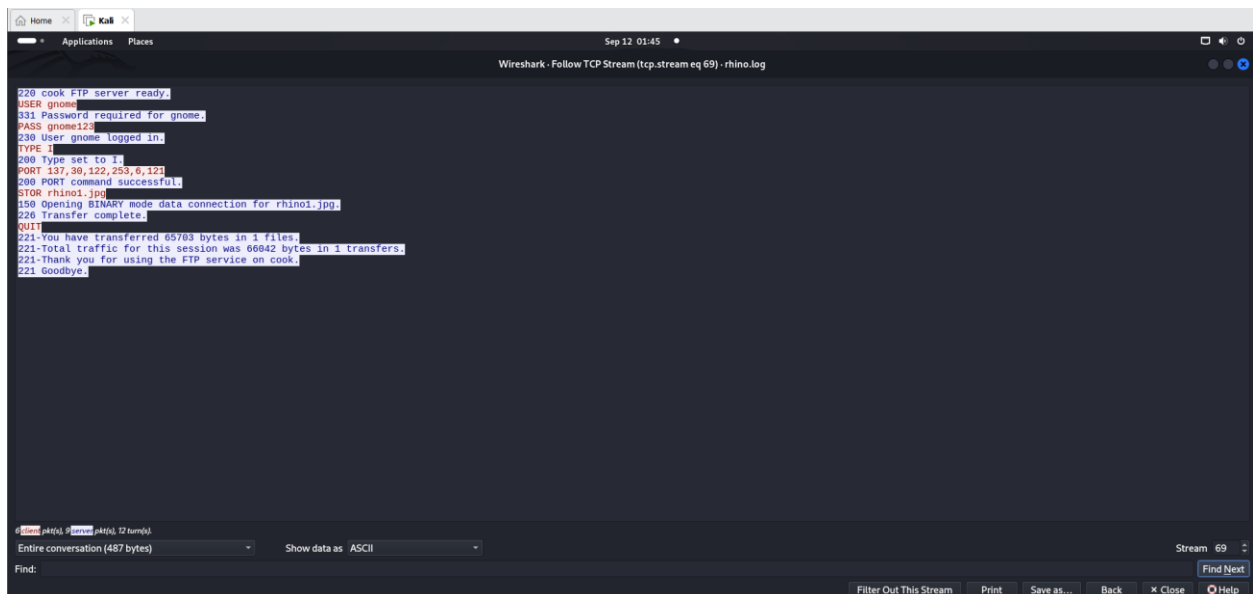
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

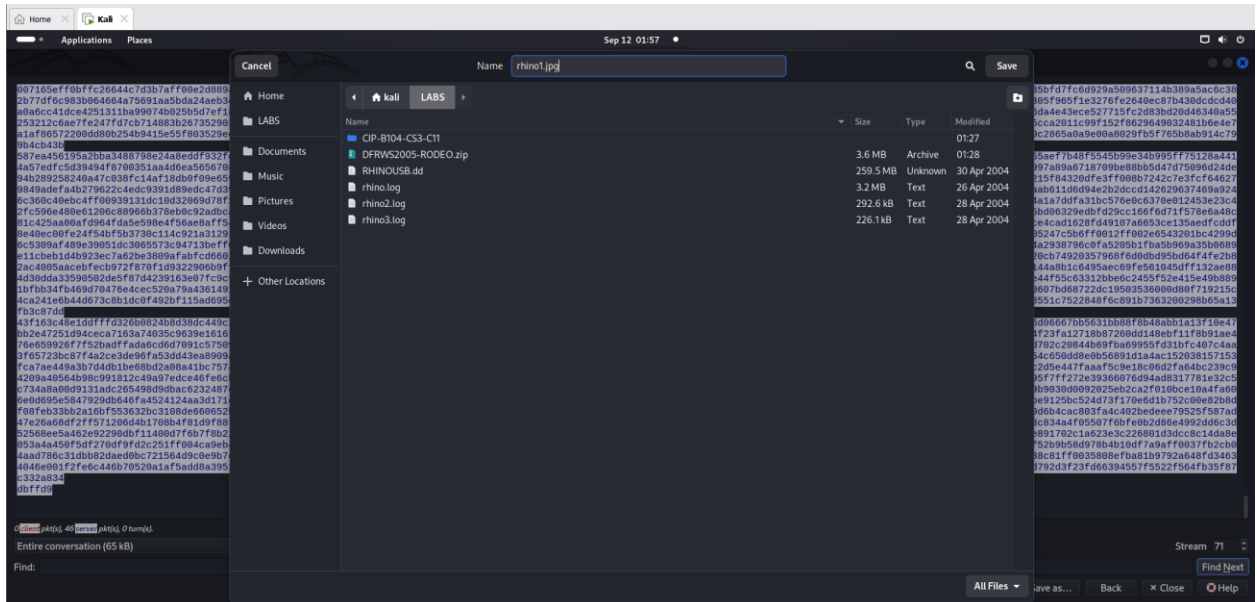
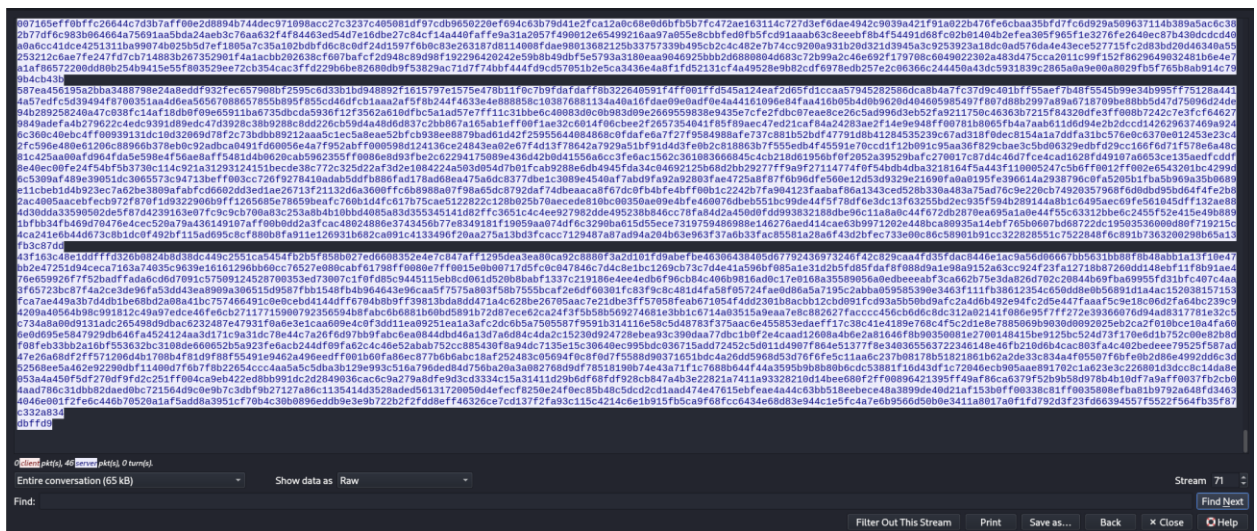
Apply a display filter ... <Ctrl-/>

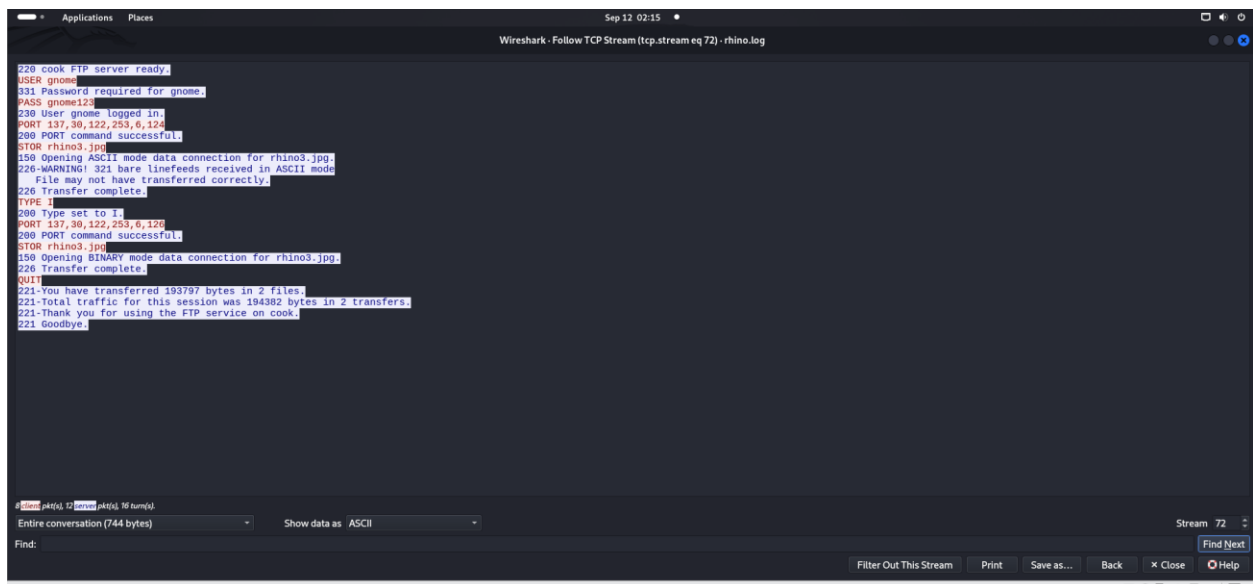
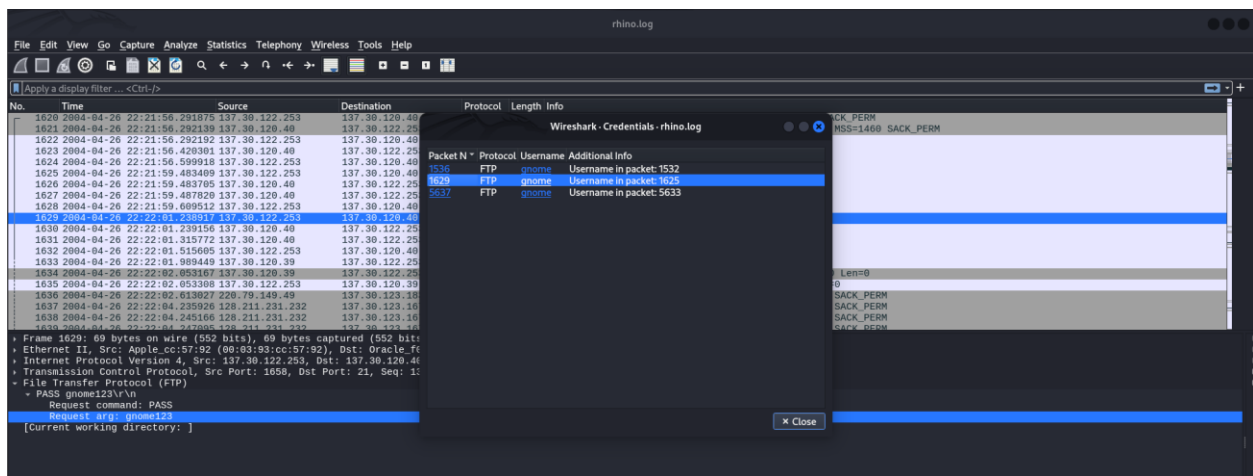
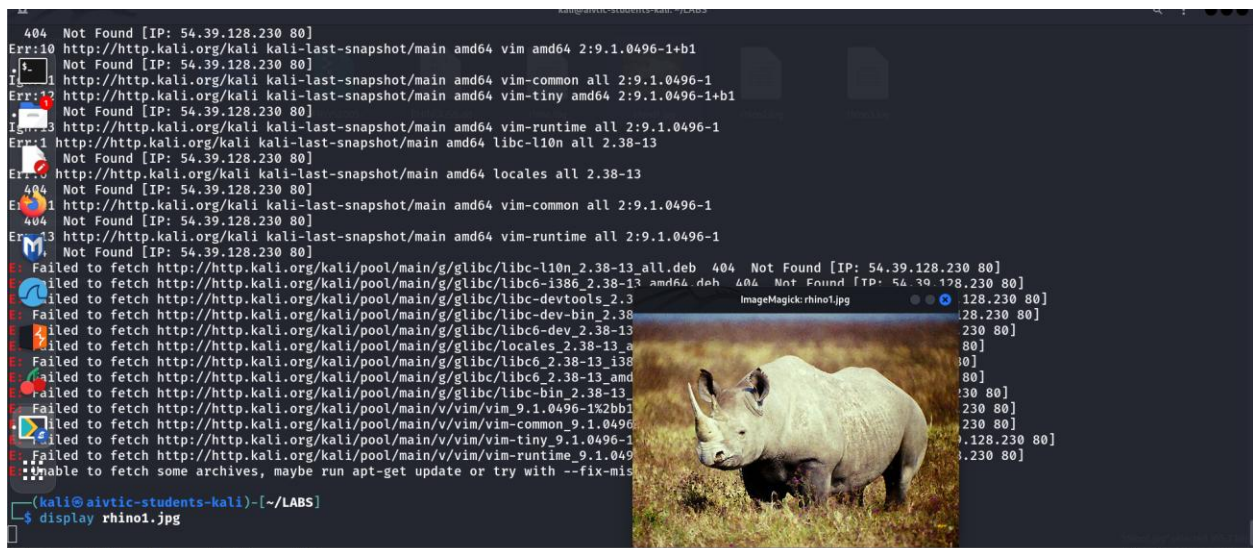
No.	Time	Source	Destination	Protocol	Length	Info
1526	2004-04-26 22:21:39.969911	137.30.122.253	137.30.120.40	TCP	62	62 1655 -> 21 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
1527	2004-04-26 22:21:39.970198	137.30.120.40	137.30.122.253	TCP	62	62 21 -> 1655 [SYN, ACK] Seq=0 Ack=1 Min=49640 Len=0 MSS=1460 SACK_PERM
1528	2004-04-26 22:21:39.970250	137.30.120.40	137.30.120.40	TCP	54	54 1655 -> 21 [ACK] Seq=1 Ack=1 Win=64240 Len=0
1529	2004-04-26 22:21:39.998180	137.30.120.40	137.30.122.253	FTP	82	82 Response: 220 cook FTP server ready.
1530	2004-04-26 22:21:40.147439	137.30.122.253	137.30.120.40	TCP	54	54 1655 -> 21 [ACK] Seq=1 Ack=29 Win=64212 Len=0
1531	2004-04-26 22:21:43.596019	137.30.122.253	137.30.120.40	TCP	62	62 4075 -> 2245 [SYN] Seq=0 Win=16384 Len=0 MSS=1460 SACK_PERM
1532	2004-04-26 22:21:43.596613	137.30.122.253	137.30.120.40	FTP	66	66 Request: USER gnome
1533	2004-04-26 22:21:43.598613	137.30.120.40	137.30.122.253	TCP	60	60 21 -> 1655 [ACK] Seq=29 Ack=13 Win=49640 Len=0
1534	2004-04-26 22:21:43.602936	137.30.120.40	137.30.122.253	FTP	88	88 Response: 331 Password required for gnome.
1535	2004-04-26 22:21:43.758957	137.30.122.253	137.30.120.40	TCP	54	54 1655 -> 21 [ACK] Seq=13 Ack=63 Win=64178 Len=0
1536	2004-04-26 22:21:45.025720	137.30.122.253	137.30.120.40	FTP	60	60 Request: PASS gnome123
1537	2004-04-26 22:21:45.025918	137.30.120.40	137.30.122.253	TCP	60	60 21 -> 1655 [ACK] Seq=63 Ack=20 Win=49640 Len=0
1538	2004-04-26 22:21:45.760912	137.30.120.40	137.30.122.253	FTP	81	81 Response: 230 User gnome logged in.
1539	2004-04-26 22:21:45.865674	137.30.122.253	137.30.120.40	TCP	54	54 1655 -> 21 [ACK] Seq=28 Ack=90 Win=64151 Len=0
1540	2004-04-26 22:21:46.569519	137.30.122.253	137.30.120.40	FTP	62	62 Request: TYPE I
1541	2004-04-26 22:21:46.569784	137.30.120.40	137.30.122.253	FTP	74	74 Response: 200 Type set to I.
1542	2004-04-26 22:21:46.668224	137.30.122.253	137.30.120.40	TCP	54	54 1655 -> 21 [ACK] Seq=36 Ack=110 Win=64131 Len=0
1543	2004-04-26 22:21:46.680481	120.130.102.117	137.30.120.40	TCP	62	62 RTP Retransmission 4075 Seq=2245 [SYN] Seq=0 Win=16384 Len=0 MSS=1460 SACK_PERM
1544	2004-04-26 22:21:49.952000	137.30.120.40	137.30.122.253	FTP	81	81 Request: PORT 137.30.122.253,6,123
1545	2004-04-26 22:21:49.953485	137.30.120.40	137.30.122.253	TCP	84	84 Response: 200 PORT command successful.
1546	2004-04-26 22:21:49.954047	137.30.122.253	137.30.120.40	FTP	71	71 Request: STOR rhino1.jpg
1547	2004-04-26 22:21:49.954833	137.30.120.40	137.30.122.253	TCP	62	62 20 -> 1657 [SYN] Seq=0 Win=49640 Len=0 MSS=1460 SACK_PERM

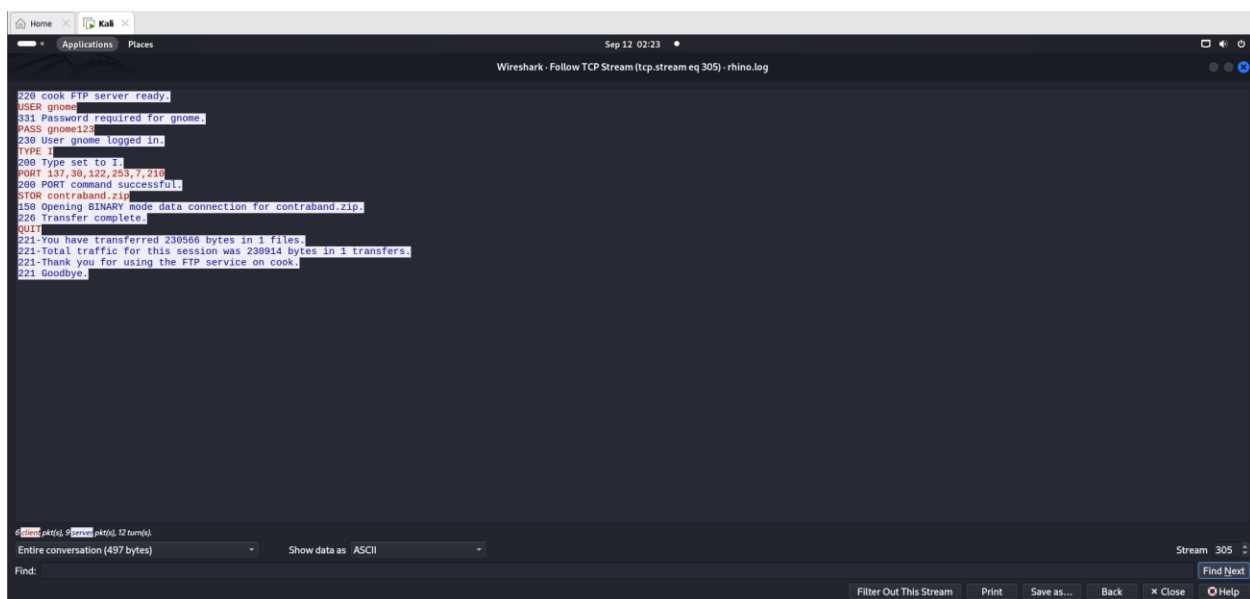
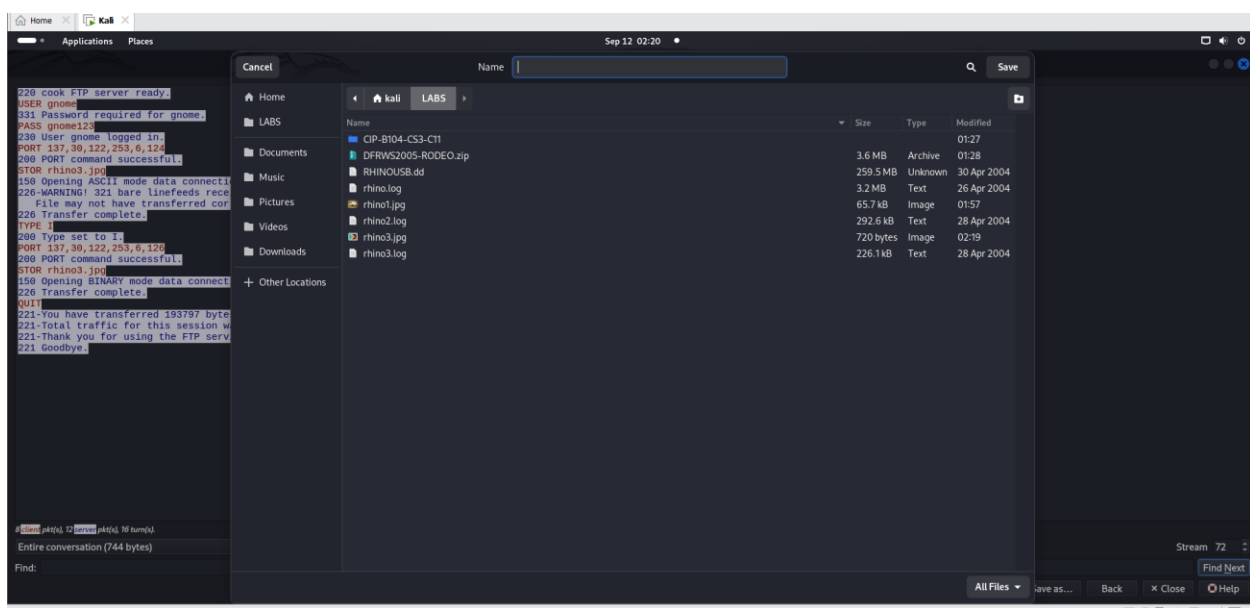
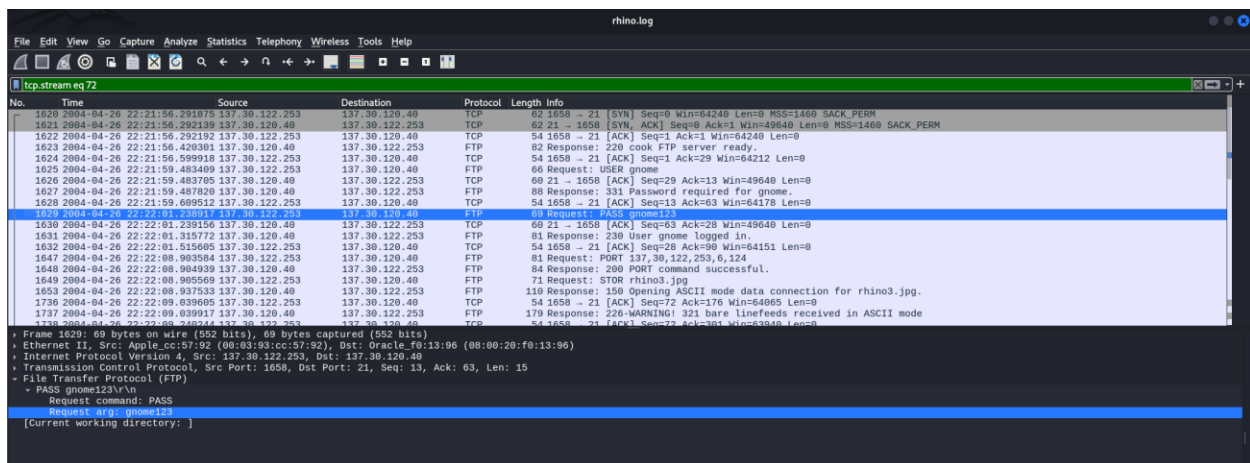
Frame 1536: 60 bytes on wire (552 bits), 60 bytes captured (552 bits) on Ethernet II, Src: Apple_cc:57:92 (08:03:93:cc:57:92), Dst: Oracle_fc:13:96 (08:00:20:fe:13:96) Internet Protocol Version 4, Src: 137.30.122.253, Dst: 137.30.120.40, Transmission Control Protocol, Src Port: 1655, Dst Port: 21, Seq: 13, Ack: 63, Len: 15 File Transfer Protocol (FTP) - PASS gnome123\r\n Request command: PASS Request arg: gnome123 [Current working directory:]

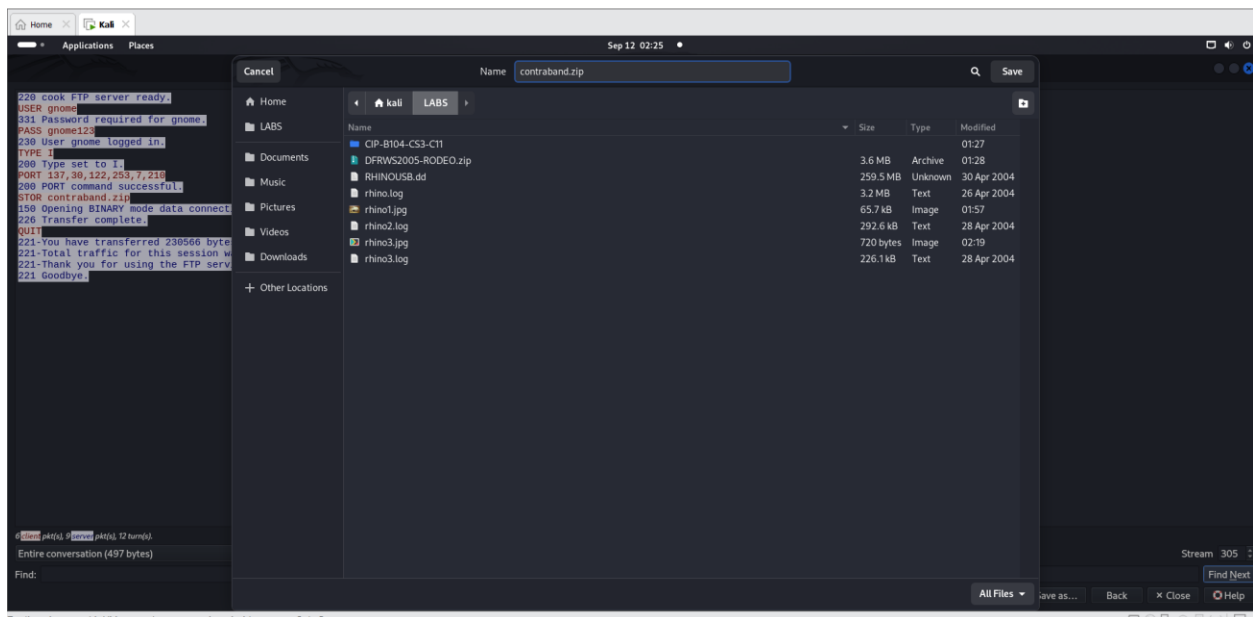
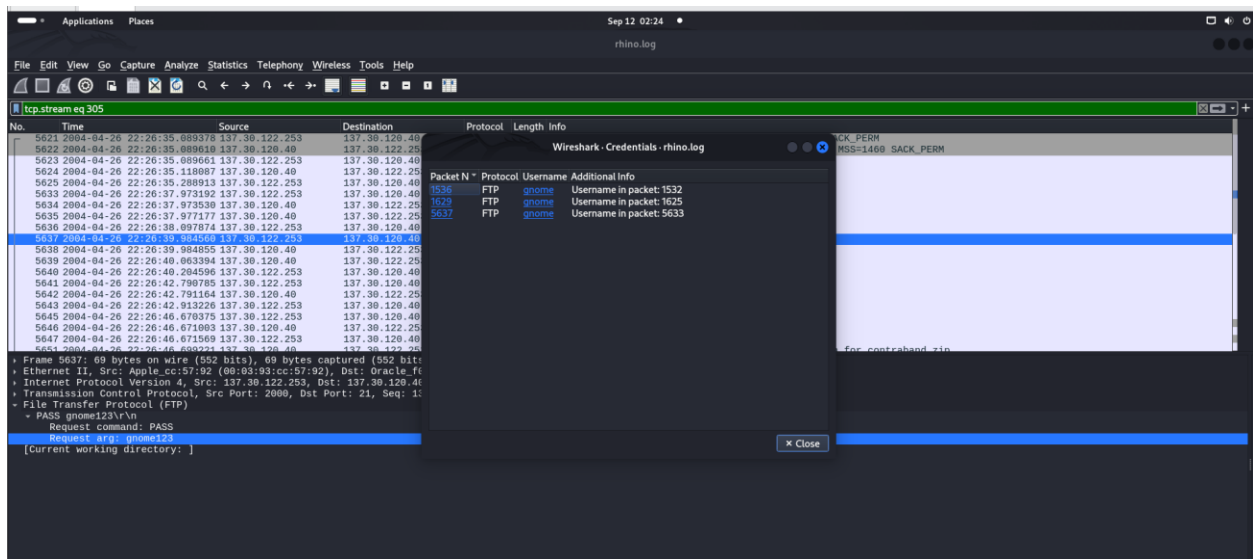
Mark/Unmark Packet(s) Ctrl+M
Ignore/Unignore Packet(s) Ctrl+D
Set/Unset Time Reference Ctrl+T
Time Shift... Ctrl+Shift+T
Packet Comments
Edit Resolved Name
Apply as Filter
Prepare as Filter
Conversation Filter :20:fe:13:96
Colorize Conversation
SCTP
Follow
Copy
Protocol Preferences
Decode As...
Show Packet in New Window



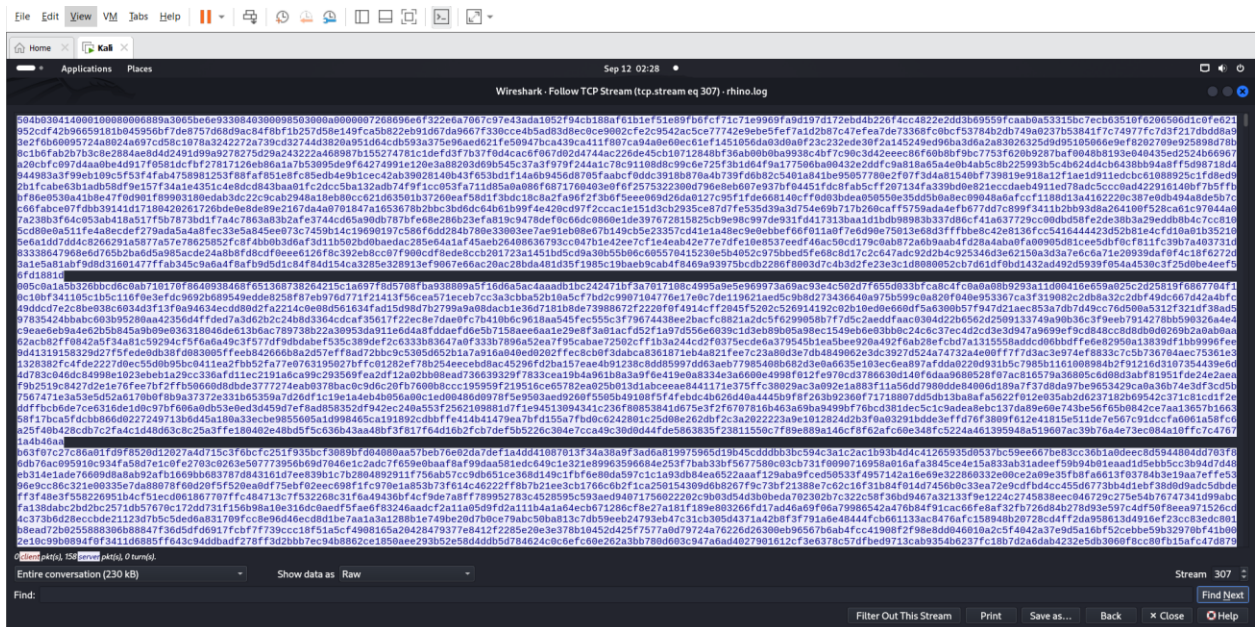
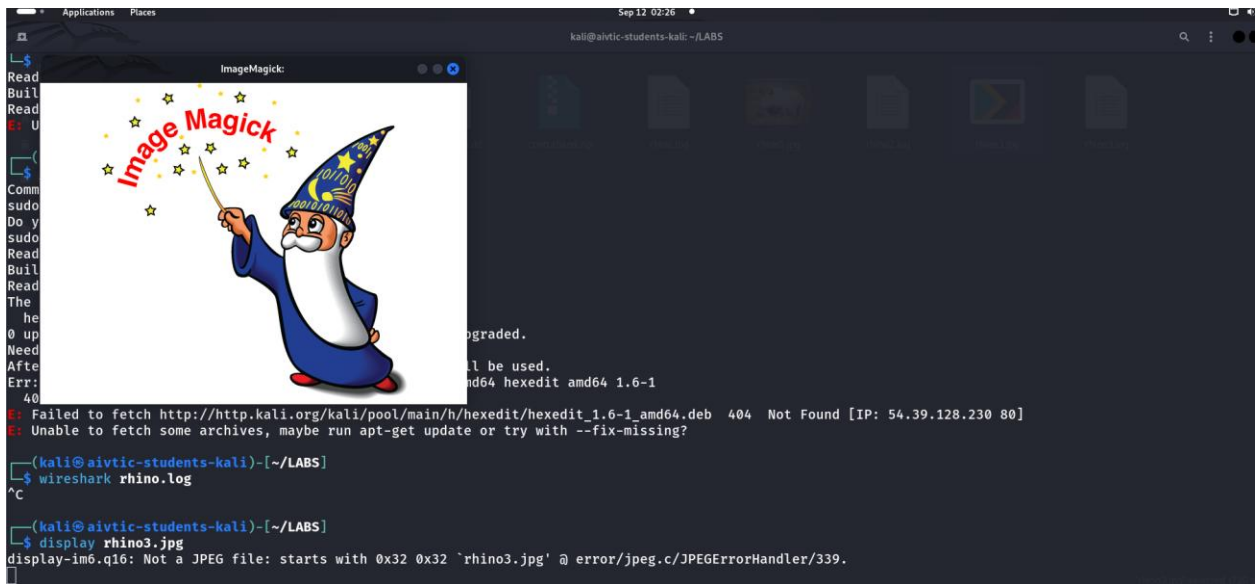


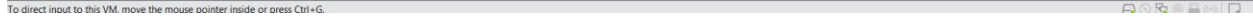






To direct input to this VM, mouse the mouse pointer inside or press Ctrl+G.





```
(kali@aivtic-students-kali)-[~/LABS]
$ mmls RHINOUSB.dd
(kali@aivtic-students-kali)-[~/LABS]
$ mmls RHINOUSB.dd
(kali@aivtic-students-kali)-[~/LABS]
$ fsstat -b 512 -o 0 RHINOUSB.dd
FILE SYSTEM INFORMATION
-----
File System Type: FAT16

OEM Name: mkdosfs
Volume ID: 0x4092d9d1
Volume Label (Boot Sector):
Volume Label (Root Directory):
File System Type Label: FAT16

Sectors before file system: 0

File System Layout (in sectors)
Total Range: 0 - 506847
* Reserved: 0 - 0
** Boot Sector: 0
* FAT 0: 1 - 248
* FAT 1: 249 - 496
```

```
Home Applications Places
Sep 12 05:03
kali@aivtic-students-kali: ~/LABS

** Boot Sector: v
* FAT 0: 1 - 248
* FAT 1: 249 - 496
* Data Area: 497 - 506847
** Root Directory: 497 - 528
** Cluster Area: 529 - 506840
** Non-clustered: 506841 - 506847

METADATA INFORMATION
-----
Range: 2 - 8101622
Root Directory: 2

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 4096
Total Cluster Range: 2 - 63290

FAT CONTENTS (in sectors)
-----
529-536 (8) -> EOF
537-544 (8) -> EOF
```

```
kali@aivtic-students-kali: ~/LABS

(kali@aivtic-students-kali)-[~/LABS]
$ icat "$CASE/working/RHINOUSB_working.dd" <METADATA_ADDRESS> \
> "$CASE/recovered/GUMBO1.TXT"
zsh: parse error near `>'

(kali@aivtic-students-kali)-[~/LABS]
$ fls -o 0 RHINOUSB.dd
r/r 4: gumbo1.txt
r/r 6: gumbo2.txt
v/v 8101619: $MBR
v/v 8101620: $FAT1
v/v 8101621: $FAT2
V/V 8101622: $OrphanFiles

(kali@aivtic-students-kali)-[~/LABS]
$ fls
Missing image name
usage: fls [-adDFlhpruvV] [-f fstype] [-i imgtype] [-b dev_sector_size] [-m dir/] [-o imgoffset] [-z ZONE] [-s seconds] image [images]
[inode]
If [inode] is not given, the root directory is used
-a: Display "." and ".." entries
-d: Display deleted entries only
-D: Display only directories
-F: Display only files
-l: Display long version (like ls -l)
-i imgtype: Format of image file (use '-i list' for supported types)
-b dev_sector_size: The size (in bytes) of the device sectors
```

```

-n: Include md5 checksum hash in mactime output
-o imgoffset: Offset into image file (in sectors)
-P pooltype: Pool container type (use '-P list' for supported types)
-B pool_volume_block: Starting block (for pool volumes only)
-S snap_id: Snapshot ID (for APFS only)
-p: Display full path for each file
-r: Recurse on directory entries
-u: Display undeleted entries only
-v: verbose output to stderr
-V: Print version
-z: Time zone of original machine (i.e. EST5EDT or GMT) (only useful with -l)
-s seconds: Time skew of original machine (in seconds) (only useful with -l & -m)
-k password: Decryption password for encrypted volumes

(kali@aivtic-students-kali)-[~/LABS]
$ icat -o 0 RHINOUSB_My_COPY.dd 4 > gumbo1_output.txt
Error stat(ing) image file (raw_open: image "RHINOUSB_My_COPY.dd" - No such file or directory)

(kali@aivtic-students-kali)-[~/LABS]
$ icat -o 0 RHINOUSB_My_COPY.dd 6 > gumbo2_output.txt
Error stat(ing) image file (raw_open: image "RHINOUSB_My_COPY.dd" - No such file or directory)

(kali@aivtic-students-kali)-[~/LABS]
$ ls -l *.txt
-rw-r--r-- 1 kali kali 0 Sep 12 05:06 gumbo1_output.txt
-rw-r--r-- 1 kali kali 0 Sep 12 05:06 gumbo2_output.txt

```

```

(kali@aivtic-students-kali)-[~/LABS]
$ cat gumbo1 output.txt | head -n 5
cat: gumbo1: No such file or directory
cat: output.txt: No such file or directory

(kali@aivtic-students-kali)-[~/LABS]
$ cat gumbo2 output.txt | head -n 5
cat: gumbo2: No such file or directory
cat: output.txt: No such file or directory

(kali@aivtic-students-kali)-[~/LABS]
$ openssl dgst -md5 RHINOUSB.dd
MD5(RHINOUSB.dd)= 80348c58eec4c328ef1f7709adc56a54

(kali@aivtic-students-kali)-[~/LABS]
$ dd if=RHINOUSB.dd of=RHINOUSB_My_COPY.dd
506848+0 records in
506848+0 records out
259506176 bytes (260 MB, 247 MiB) copied, 1.04566 s, 248 MB/s

(kali@aivtic-students-kali)-[~/LABS]
$ openssl dgst -md5 RHINOUSB_My_COPY.dd
MD5(RHINOUSB_My_COPY.dd)= 80348c58eec4c328ef1f7709adc56a54

(kali@aivtic-students-kali)-[~/LABS]
$ photorec RHINOUSB_My_COPY.dd

```

```

MD5(RHINOUSB_My_COPY.dd)= 80348c58eec4c328ef1f7709adc56a54

(kali@aivtic-students-kali)-[~/LABS]
$ photorec RHINOUSB_My_COPY.dd
PhotoRec 7.1, Data Recovery Utility, July 2019
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

(kali@aivtic-students-kali)-[~/LABS]
$

```

```
PhotoRec 7.1, Data Recovery Utility, July 2019
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

PhotoRec is free software, and
comes with ABSOLUTELY NO WARRANTY.

Select a media (use Arrow keys, then press Enter):
>Disk RHINOUSB_My_COPY.dd - 259 MB / 247 MiB (RO)

>[Proceed ] [ Sudo ] [ Quit ]

Note: Some disks won't appear unless you're root user.
Disk capacity must be correctly detected for a successful recovery.
If a disk listed above has an incorrect size, check HD jumper settings and BIOS
detection, and install the latest OS patches and disk drivers.
```

```
PhotoRec 7.1, Data Recovery Utility, July 2019
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk RHINOUSB_My_COPY.dd - 259 MB / 247 MiB (RO)

Partition      Start      End      Size in sectors
Unknown        0  0  1  1021  6 60  506848 [Whole disk]
> P FAT16       0  0  1  1021  6 60  506848

>[ Search ] [Options ] [File Opt] [ Quit ]
Start file recovery
```

```
PhotoRec 7.1, Data Recovery Utility, July 2019
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

P FAT16          0  0  1 1021  6 60    506848

To recover lost files, PhotoRec needs to know the filesystem type where the
file were stored:
[ ext2/ext3 ] ext2/ext3/ext4 filesystem
>[ Other     ] FAT/NTFS/HFS+/ReiserFS/...
```

```
PhotoRec 7.1, Data Recovery Utility, July 2019
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

P FAT16          0  0  1 1021  6 60    506848

Please choose if all space needs to be analysed:
>[ Free       ] Scan for files from FAT16 unallocated space only
[ Whole      ] Extract files from whole partition
```

```
Applications  Places
Sep 12 05:18
kali@aivtic-students-kali: ~/LABS

PhotoRec 7.1, Data Recovery Utility, July 2019
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk RHINOUSB_My_COPY.dd - 259 MB / 247 MiB (RO)
Partition      Start      End      Size in sectors
P FAT16         0    0 1 1021    6 60    506848

132 files saved in /home/kali/LABS/recup_dir directory.
Recovery completed.

You are welcome to donate to support and encourage further development
https://www.cgsecurity.org/wiki/Donation

[ Quit ]
```

```
(kali@aivtic-students-kali)-[~/LABS]
$ ls -l
total 514144
drwxr-xr-x 3 kali kali 4096 Sep 12 01:27 CIP-B104-CS3-C11
-rw-r--r-- 1 kali kali 3614418 Sep 12 01:28 DFRWS2005-RODEO.zip
-rw-r--r-- 1 kali kali 259506176 Apr 30 2004 RHINOUSB.dd
-rw-r--r-- 1 kali kali 259506176 Sep 12 05:08 RHINOUSB_My_COPY.dd
-rw-r--r-- 1 kali kali 480 Sep 12 02:25 contraband.zip
-rw-r--r-- 1 kali kali 0 Sep 12 05:06 gumbo1_output.txt
-rw-r--r-- 1 kali kali 0 Sep 12 05:06 gumbo2_output.txt
-rw-r--r-- 1 kali kali 40960 Sep 12 05:14 photorec.se2
drwxr-xr-x 2 kali kali 4096 Sep 12 05:17 recup_dir.1
drwxr-xr-x 2 kali kali 4096 Sep 12 05:18 recup_dir.2
-rw-r--r-- 1 kali kali 3187907 Apr 26 2004 rhino.log
-rw-r--r-- 1 kali kali 65703 Sep 12 01:57 rhino1.jpg
-rw-r--r-- 1 kali kali 292604 Apr 28 2004 rhino2.log
-rw-r--r-- 1 kali kali 720 Sep 12 02:19 rhino3.jpg
-rw-r--r-- 1 kali kali 226094 Apr 28 2004 rhino3.log
```

```
(kali@aivtic-students-kali)-[~/LABS]
$ ls recup_dir.1/
f0000545.txt f0339025.txt f0348793.txt f0358561.txt f0368329.txt f0378097.txt f0387865.txt
f0104057.jpg f0339545.txt f0349313.txt f0359073.txt f0368841.txt f0378609.txt f0388377.txt
f0104249.jpg f0340057.txt f0349825.txt f0359593.txt f0369361.txt f0379129.txt f0388889.txt
f0105005.jpg f0340569.txt f0350337.txt f0360105.txt f0369873.txt f0379641.txt f0389409.txt
f0105873.jpg f0341081.txt f0350849.txt f0360969.txt f0370385.txt f0380153.txt f0389921.txt
f0106393.jpg f0341601.txt f0351369.txt f0361137.txt f0370897.txt f0380665.txt f0390433.txt
f0106409.jpg f0342113.txt f0351881.txt f0361649.txt f0371417.txt f0381185.txt f0390945.txt
f0106805.gif f0342625.txt f0352393.txt f0362161.txt f0371929.txt f0381697.txt f0391465.txt
f0106889.gif f0343137.txt f0352905.txt f0362673.txt f0372441.txt f0382209.txt f0391977.txt
f0106905.txt f0343657.txt f0353425.txt f0363193.txt f0372953.txt f0382721.txt f0392489.txt
f0335017_She_died_in_February_at_the_age_of_74.doc f0344585.txt f0353937.txt f0363705.txt f0373473.txt f0383241.txt f0393001.txt
f0335081.jpg f0344681.txt f0354449.txt f0364217.txt f0373985.txt f0383753.txt f0393737.txt
f0335601.txt f0345201.txt f0354961.txt f0364729.txt f0374497.txt f0384265.txt f0394033.txt
f0335945.txt f0345713.txt f0355481.txt f0365249.txt f0375009.txt f0384777.txt f0394545.txt
f0336457.txt f0346225.txt f0355993.txt f0365761.txt f0375529.txt f0385297.txt f0395065.txt
f0336969.txt f0346737.txt f0356505.txt f0366273.txt f0376041.txt f0385809.txt f0395577.txt
f0337489.txt f0347257.txt f0357017.txt f0366785.txt f0376553.txt f0386321.txt f0396089.txt
f0338001.txt f0347769.txt f0357537.txt f0367305.txt f0377353.txt f0386833.txt f0396601.txt
f0338513.txt f0348281.txt f0358049.txt f0367817.txt f0377585.txt f0387353.txt report.xml
```

```
(kali@aivtic-students-kali)-[~/LABS]
$ cd recup_dir.1/

(kali@aivtic-students-kali)-[~/LABS/recup_dir.1]
$ ls -l *.jpg
-rw-r--r-- 1 kali kali 95814 Sep 12 05:17 f0104057.jpg
-rw-r--r-- 1 kali kali 415534 Sep 12 05:17 f0104249.jpg
-rw-r--r-- 1 kali kali 411361 Sep 12 05:17 f0105065.jpg
-rw-r--r-- 1 kali kali 264600 Sep 12 05:17 f0105873.jpg
-rw-r--r-- 1 kali kali 6809 Sep 12 05:17 f0106393.jpg
-rw-r--r-- 1 kali kali 230665 Sep 12 05:17 f0106409.jpg
-rw-r--r-- 1 kali kali 264600 Sep 12 05:17 f0335081.jpg

(kali@aivtic-students-kali)-[~/LABS/recup_dir.1]
$
```

```
kali@aivtic-students-kali: ~/LABS/recup_dir.1
f0335017_She_died_in_February_at_the_age_of_74.doc f0344585.txt f0353937.txt f0363705.txt f0373473.txt f0383241.txt f0393001.txt
f0335081.jpg f0344681.txt f0354449.txt f0364217.txt f0373985.txt f0383753.txt f0393737.txt
f03350601.txt f0345201.txt f0354961.txt f0364217.txt f0373985.txt f0383753.txt f0393737.txt
f0335945.txt f0345713.txt f0355481.txt f0364217.txt f0373985.txt f0383753.txt f0393737.txt
f0335945.txt f0346225.txt f0355993.txt f0364217.txt f0373985.txt f0383753.txt f0393737.txt
f0336969.txt f0346737.txt f0356505.txt f0364217.txt f0373985.txt f0383753.txt f0393737.txt
f0337489.txt f0347257.txt f0357017.txt f0364217.txt f0373985.txt f0383753.txt f0393737.txt
f0338001.txt f0347769.txt f0357537.txt f0364217.txt f0373985.txt f0383753.txt f0393737.txt
f0338513.txt f0348281.txt f0358049.txt f0364217.txt f0373985.txt f0383753.txt f0393737.txt
report.xml

(kali@aivtic-students-kali)-[~/LABS]
$ cd recup_dir.1/

(kali@aivtic-students-kali)-[~/LABS/recup_dir.1]
$ ls -l *.jpg
-rw-r--r-- 1 kali kali 95814 Sep 12 05:17 f0104057.jpg
-rw-r--r-- 1 kali kali 415534 Sep 12 05:17 f0104249.jpg
-rw-r--r-- 1 kali kali 411361 Sep 12 05:17 f0105065.jpg
-rw-r--r-- 1 kali kali 264600 Sep 12 05:17 f0105873.jpg
-rw-r--r-- 1 kali kali 6809 Sep 12 05:17 f0106393.jpg
-rw-r--r-- 1 kali kali 230665 Sep 12 05:17 f0106409.jpg
-rw-r--r-- 1 kali kali 264600 Sep 12 05:17 f0335081.jpg

(kali@aivtic-students-kali)-[~/LABS/recup_dir.1]
$ display f0104057.jpg
```



```
kali@aivtic-students-kali: ~/LABS/recup_dir.1
f0335017_She_died_in_February_at_the_age_of_74.doc f0344585.txt f0353937.txt f0363705.txt f0373473.txt f0383241.txt f0393001.txt
f0335081.jpg f0344681.txt f0354449.txt f0364217.txt f0373985.txt f0383753.txt f0393737.txt
f03350601.txt f0345201.txt f0354961.txt f0364217.txt f0373985.txt f0383753.txt f0393737.txt
f0335945.txt f0345713.txt f0355481.txt f0364217.txt f0373985.txt f0383753.txt f0393737.txt
f0335945.txt f0346225.txt f0355993.txt f0364217.txt f0373985.txt f0383753.txt f0393737.txt
f0336969.txt f0346737.txt f0356505.txt f0364217.txt f0373985.txt f0383753.txt f0393737.txt
f0337489.txt f0347257.txt f0357017.txt f0364217.txt f0373985.txt f0383753.txt f0393737.txt
f0338001.txt f0347769.txt f0357537.txt f0364217.txt f0373985.txt f0383753.txt f0393737.txt
f0338513.txt f0348281.txt f0358049.txt f0364217.txt f0373985.txt f0383753.txt f0393737.txt
report.xml

(kali@aivtic-students-kali)-[~/LABS/recup_dir.1]
$ display f0104057.jpg

(kali@aivtic-students-kali)-[~/LABS/recup_dir.1]
$ display f0105067.jpg
display-im6.q16: unable to open image `f0105067.jpg': No such file or directory @ error/blob.c/OpenBlob/2964.
```



Steganography

Illegal possession investigations

```
(kali@aivtic-students-kali)-[~/LABS]
$ mkdir steg

(kali@aivtic-students-kali)-[~/LABS]
$ cp recup_dir.1/f0104057.jpg steg

(kali@aivtic-students-kali)-[~/LABS]
$ cp recup_dir.1/f0104249.jpg steg/

(kali@aivtic-students-kali)-[~/LABS]
$ cp recup_dir.1/f0105065.jpg steg

(kali@aivtic-students-kali)-[~/LABS]
$ cp recup_dir.1/f0105873.jpg steg

(kali@aivtic-students-kali)-[~/LABS]
$ cp recup_dir.1/f0335081.jpg steg

(kali@aivtic-students-kali)-[~/LABS]
$ cd steg
```

```
(kali@aivtic-students-kali)-[~/LABS]
$ cd steg

(kali@aivtic-students-kali)-[~/LABS/steg]
$ ls
f0104057.jpg  f0105065.jpg  f0335081.jpg
f0104249.jpg  f0105873.jpg

(kali@aivtic-students-kali)-[~/LABS/steg]
$
```

```
File Edit View Settings Tools Window Help
Kali
Applications Places
Sep 12 05:34
kali@aivtic-students-kali: ~/LABS/steg

$ sudo apt install exiftool
[sudo] password for kali:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Note, selecting 'libimage-exiftool-perl' instead of 'exiftool'
libimage-exiftool-perl is already the newest version (12.76+dfsg-1).
0 upgraded, 0 newly installed, 0 to remove and 2382 not upgraded.

(kali@aivtic-students-kali)-[~/LABS/steg]
$ exiftool f0104057.jpg
ExifTool Version Number      : 12.76
File Name                    : f0104057.jpg
Directory                   : .
File Size                    : 96 kB
File Modification Date/Time  : 2026:09:12 05:27:34-04:00
File Access Date/Time       : 2026:09:12 05:27:34-04:00
File Inode Change Date/Time  : 2026:09:12 05:27:34-04:00
File Permissions             : -rw-r--r--
File Type                    : JPEG
File Type Extension         : jpg
MIME Type                    : image/jpeg
JFIF Version                 : 1.01
```

```
File Access Date/Time      : 2026:09:12 05:27:34-04:00
File Inode Change Date/Time : 2026:09:12 05:27:34-04:00
File Permissions          : -rw-r--r--
File Type                 : JPEG
File Type Extension       : jpg
MIME Type                  : image/jpeg
JFIF Version              : 1.01
Resolution Unit           : None
X Resolution               : 0
Y Resolution               : 0
Comment                   : copyright 2000 philg@mit.edu
Image Width                : 528
Image Height               : 792
Encoding Process           : Baseline DCT, Huffman coding
Bits Per Sample            : 8
Color Components           : 3
Y Cb Cr Sub Sampling       : YCbCr4:4:4 (1 1)
Image Size                 : 528x792
Megapixels                 : 0.418

(kali@aivtic-students-kali)-[~/LABS/steg]
$
```

```
, 372.01 KiB | 563.00Receiving objects: 68% (285/419), 372.01 KiB | 563.00Receiving objects: 69% (290/419), 3
72.01 KiB | 563.00Receiving objects: 70% (294/419), 372.01 KiB | 563.00Receiving objects: 71% (298/419), 372.
01 KiB | 563.00Receiving objects: 72% (302/419), 372.01 KiB | 563.00Receiving objects: 73% (306/419), 372.01
KiB | 563.00Receiving objects: 74% (311/419), 372.01 KiB | 563.00Receiving objects: 75% (315/419), 372.01 KiB
| 563.00Receiving objects: 76% (319/419), 372.01 KiB | 563.00Receiving objects: 77% (323/419), 940.01 KiB |
727.00Receiving objects: 78% (327/419), 940.01 KiB | 727.00Receiving objects: 79% (332/419), 940.01 KiB | 727
.00Receiving objects: 80% (336/419), 940.01 KiB | 727.00Receiving objects: 81% (340/419), 940.01 KiB | 727.00
Receiving objects: 82% (344/419), 940.01 KiB | 727.00Receiving objects: 83% (348/419), 940.01 KiB | 727.00Rec
eiving objects: 84% (352/419), 940.01 KiB | 727.00Receiving objects: 85% (357/419), 940.01 KiB | 727.00Receiv
ing objects: 86% (361/419), 940.01 KiB | 727.00Receiving objects: 87% (365/419), 940.01 KiB | 727.00Receiving
objects: 88% (369/419), 940.01 KiB | 727.00Receiving objects: 89% (373/419), 1.25 MiB | 707.00 KReceiving ob
jects: 90% (378/419), 1.25 MiB | 707.00 KReceiving objects: 91% (382/419), 1.25 MiB | 707.00 KReceiving objec
ts: 92% (386/419), 1.25 MiB | 707.00 KReceiving objects: 93% (390/419), 1.25 MiB | 707.00 KReceiving objects:
94% (394/419), 1.25 MiB | 707.00 KReceiving objects: 95% (399/419), 1.25 MiB | 707.00 KReceiving objects: 9
6% (403/419), 1.25 MiB | 707.00 KReceiving objects: 97% (407/419), 1.25 MiB | 707.00 KReceiving objects: 98%
(411/419), 1.25 MiB | 707.00 Kremote: Total 419 (delta 0), reused 0 (delta 0), pack-reused 419 (from 1)
Receiving objects: 99% (415/419), 1.25 MiB | 707.00 KReceiving objects: 100% (419/419), 1.25 MiB | 707.00 KRec
eiving objects: 100% (419/419), 1.38 MiB | 655.00 KiB/s, done.
Resolving deltas: 100% (59/59), done.

(kali@aivtic-students-kali)-[~/LABS/steg]
$
```

```
kali@kali-students-kali: ~/LABS/steg/stegdetect
autoheader: WARNING: and 'config.h.top', to define templates for 'config.h.in'
autoheader: WARNING: is deprecated and discouraged.
autoheader:
autoheader: WARNING: Using the third argument of 'AC_DEFINE_UNQUOTED' and
autoheader: WARNING: 'AC_DEFINE' allows one to define a template without
autoheader: WARNING: 'acconfig.h':
autoheader:
autoheader: WARNING: AC_DEFINE([NEED_FUNC_MAIN], 1,
autoheader: WARNING: [Define if a function 'main' is needed.])
autoheader:
autoheader: WARNING: More sophisticated templates can also be produced, see the
autoheader: WARNING: documentation.
automake: warning: autoconf input should be named 'configure.ac', not 'configure.in'
configure.in:6: warning: AM_INIT_AUTOMAKE: two- and three-arguments forms are deprecated. For more info, see:
configure.in:6: https://www.gnu.org/software/automake/manual/automake.html#Modernize-AM_005fINIT_005fAUTOMAKE-i
nvocation
configure.in:15: installing './compile'
Makefile.am:30: warning: 'INCLUDES' is the old name for 'AM_CPPFLAGS' (or '*_CPPFLAGS')
automake: warning: autoconf input should be named 'configure.ac', not 'configure.in'
(kali@aivtic-students-kali)-[~/LABS/steg/stegdetect]
$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

```
smatch]
172 |         memset(ctx, 0, sizeof(ctx)); /* In case it's sensitive */
    |         ^~~~~~
md5.c:172:9: note: include '<string.h>' or provide a declaration of 'memset'
md5.c:172:30: warning: argument to 'sizeof' in 'memset' call is the same expression as the destination; did you
mean to dereference it? [-Wsizeof-pointer-memaccess]
172 |         memset(ctx, 0, sizeof(ctx)); /* In case it's sensitive */
    |         ^~~~~~
gcc -O2 -Wall -g -o stegdetect common.o jphide_table.o util.o stegdetect.o chi2cdf.o extraction.o discrimina
tion.o math.o dct.o jutil.o f5.o strlcat.o strlcpy.o dirname.o md5.o -L./jpeg-6b -ljpeg -L./file -lfile -lm -L.
/jpeg-6b -ljpeg
/usr/bin/ld: ./file/libfile.a(file.o):/home/kali/LABS/steg/stegdetect/file/file.c:96: multiple definition of `p
rogramname'; stegdetect.o:/home/kali/LABS/steg/stegdetect/stegdetect.c:76: first defined here
collect2: error: ld returned 1 exit status
make[2]: *** [Makefile:559: stegdetect] Error 1
make[2]: Leaving directory '/home/kali/LABS/steg/stegdetect'
make[1]: *** [Makefile:633: all-recursive] Error 1
make[1]: Leaving directory '/home/kali/LABS/steg/stegdetect'
make: *** [Makefile:450: all] Error 2
(kali@aivtic-students-kali)-[~/LABS/steg/stegdetect]
$
```

```
(kali@aivtic-students-kali)-[~/LABS/steg/stegdetect]
$ mkdir stegseekexe

(kali@aivtic-students-kali)-[~/LABS/steg/stegdetect]
$ cd stegseekexe/

(kali@aivtic-students-kali)-[~/LABS/steg/stegdetect/stegseekexe]
$ wget -q ftp://ftp.gwdg.de/pub/linux/misc/ppdd/jphs_05.zip

(kali@aivtic-students-kali)-[~/LABS/steg/stegdetect/stegseekexe]
$ ls
jphs_05.zip

(kali@aivtic-students-kali)-[~/LABS/steg/stegdetect/stegseekexe]
$
```

```
Home Applications Places Sep 12 06:02
kali@aivtic-students-kali: ~/LABS/steg/stegdetect/stegseekexe
(kali@aivtic-students-kali)-[~/LABS/steg/stegdetect/stegseekexe]
$ unzip jphs_05.zip
Archive: jphs_05.zip
extracting: jphs05.zip
extracting: jphs05.zip.sig

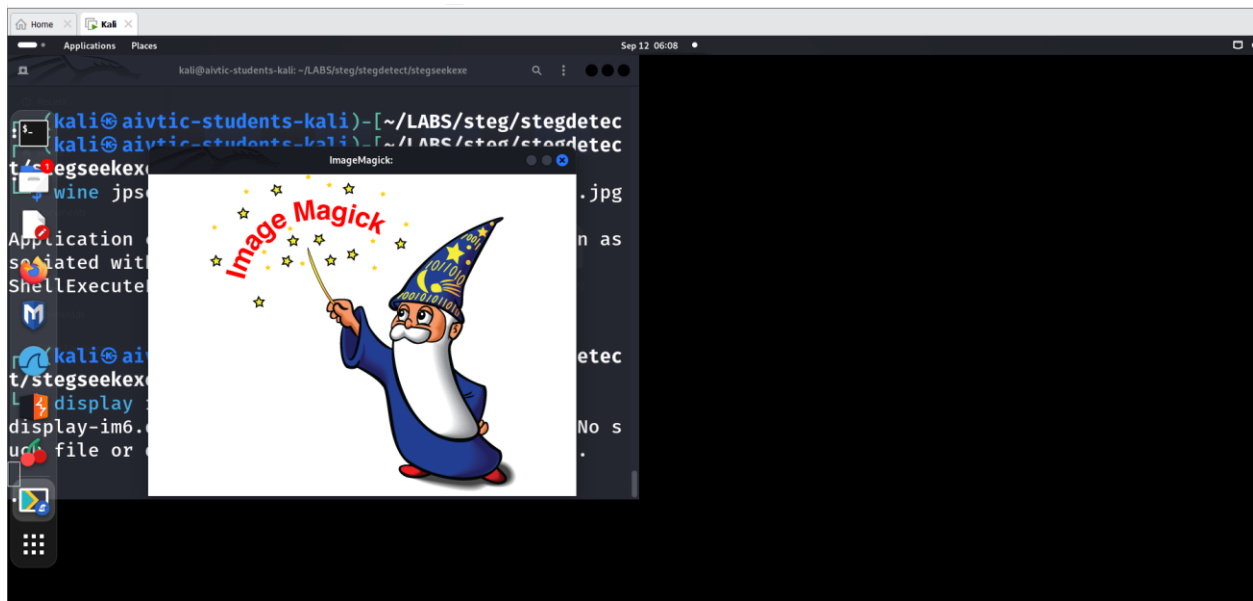
(kali@aivtic-students-kali)-[~/LABS/steg/stegdetect/stegseekexe]
$ wine jpseek.exe -/rhino/steg/f0105065.jpgr065.ino
wine: created the configuration directory '/home/
004c:err:ole:StdMarshalImpl_MarshalInterface Failed to marshal the interface {6d5140c1-7436-11ce-8034-00aa006009fa}, hr 0x80004002
0054:err:ole:StdMarshalImpl_MarshalInterface Failed to marshal the interface {6d5140c1-7436-11ce-8034-00aa006009fa}, hr 0x80004002
0054:err:ole:CoMarshalInterface Failed to marshal the interface {6d5140c1-7436-11ce-8034-00aa006009fa}, hr 0x80004002
0054:err:ole:apartment_get_local_server_stream Failed: 0x80004002
004c:err:ole:CoMarshalInterface Failed to marshal the interface {6d5140c1-7436-11ce-8034-00aa006009fa}, hr 0x80004002
004c:err:ole:apartment_get_local_server_stream Failed: 0x80004002
0054:err:ole:start_rpcss Failed to open RpcSs service
0058:err:clipboard:convert_selection Timed out waiting for SelectionNotify event
0058:err:clipboard:convert_selection Timed out waiting for SelectionNotify event
```

```
Home Applications Places Sep 12 06:06
kali@aivtic-students-kali: ~/LABS/steg/stegdetect/stegseekexe
(kali@aivtic-students-kali)-[~/LABS/steg/stegdetect/stegseekexe]
(kali@aivtic-students-kali)-[~/LABS/steg/stegdetect/stegseekexe]
$ wine jpseek.exe -/rhino/steg/f0105065.jpg r065.jpg
Application could not be started, or no application associated with the specified file.
ShellExecuteEx failed: File not found.

(kali@aivtic-students-kali)-[~/LABS/steg/stegdetect/stegseekexe]
$ display r065.jpg
display-im6.q16: unable to open image `r065.jpg': No such file or directory @ error/blob.c/OpenBlob/2964.
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.





```
kali@aivtic-students-kali: ~/LABS/steg/stegdetect/stegseekexe
ing objects: 67% (163/242), 12.71 MiB | 1.73 MiReceiving objects: 68% (165/242), 12.71 MiB | 1.73 MiReceiving
objects: 69% (167/242), 12.71 MiB | 1.73 MiReceiving objects: 70% (170/242), 12.71 MiB | 1.73 MiReceiving ob
jects: 71% (172/242), 12.71 MiB | 1.73 MiReceiving objects: 72% (175/242), 12.71 MiB | 1.73 MiReceiving objec
ts: 73% (177/242), 12.71 MiB | 1.73 MiReceiving objects: 74% (180/242), 12.71 MiB | 1.73 MiReceiving objects:
75% (182/242), 12.71 MiB | 1.73 MiReceiving objects: 76% (184/242), 12.71 MiB | 1.73 MiReceiving objects: 7
7% (187/242), 12.71 MiB | 1.73 MiReceiving objects: 78% (189/242), 12.71 MiB | 1.73 MiReceiving objects: 79%
(192/242), 12.71 MiB | 1.73 MiReceiving objects: 80% (194/242), 12.71 MiB | 1.73 MiReceiving objects: 81% (19
7/242), 12.71 MiB | 1.73 MiReceiving objects: 82% (199/242), 12.71 MiB | 1.73 MiReceiving objects: 83% (201/2
42), 12.71 MiB | 1.73 MiReceiving objects: 84% (204/242), 12.71 MiB | 1.73 MiReceiving objects: 85% (206/242)
, 12.71 MiB | 1.73 MiReceiving objects: 86% (209/242), 12.71 MiB | 1.73 MiReceiving objects: 87% (211/242), 1
2.71 MiB | 1.73 MiReceiving objects: 88% (213/242), 12.71 MiB | 1.73 MiReceiving objects: 89% (216/242), 12.7
1 MiB | 1.73 MiRemote: Total 242 (delta 0), reused 0 (delta 0), pack-reused 242 (from 1)
Receiving objects: 90% (218/242), 12.71 MiB | 1.73 MiReceiving objects: 91% (221/242), 12.71 MiB | 1.73 MiRec
eiving objects: 92% (223/242), 12.71 MiB | 1.73 MiReceiving objects: 93% (226/242), 12.71 MiB | 1.73 MiReceiv
ing objects: 94% (228/242), 12.71 MiB | 1.73 MiReceiving objects: 95% (230/242), 12.71 MiB | 1.73 MiReceiving
objects: 96% (233/242), 12.71 MiB | 1.73 MiReceiving objects: 97% (235/242), 12.71 MiB | 1.73 MiReceiving ob
jects: 98% (238/242), 12.71 MiB | 1.73 MiReceiving objects: 99% (240/242), 12.71 MiB | 1.73 MiReceiving objec
ts: 100% (242/242), 12.71 MiB | 1.73 MiReceiving objects: 100% (242/242), 13.03 MiB | 1.65 MiB/s, done.
Resolving deltas: 100% (117/117), done.

(kali@aivtic-students-kali)~/LABS/steg/stegdetect/stegseekexe
$
```

Resolving deltas: 100% (117/117), done.

```
(kali@aivtic-students-kali)~/LABS/steg/stegdetect/stegseekexe
$ ls
jphs05.zip jphs05.zip.sig jphs_05.zip stego-toolkit

(kali@aivtic-students-kali)~/LABS/steg/stegdetect/stegseekexe
$
```

Traffic Log Analysis

```
(kali@aivtic-students-kali)-[~/LABS]
$ ls
CIP-B104-CS3-C11  contraband.zip  recup_dir.1  rhino2.log
DFRWS2005-RODE0.zip  gumbo1_output.txt  recup_dir.2  rhino3.jpg
RHINOUSB.dd  gumbo2_output.txt  rhino.log  rhino3.log
RHINOUSB_My_COPY.dd  photorec.se2  rhino1.jpg  steg

(kali@aivtic-students-kali)-[~/LABS]
$ ls -l
total 514148
drwxr-xr-x 3 kali kali 4096 Sep 12 01:27 CIP-B104-CS3-C11
-rw-r--r-- 1 kali kali 3614418 Sep 12 01:28 DFRWS2005-RODE0.zip
-rw-r--r-- 1 kali kali 259506176 Apr 30 2004 RHINOUSB.dd
-rw-r--r-- 1 kali kali 259506176 Sep 12 05:08 RHINOUSB_My_COPY.dd
-rw-r--r-- 1 kali kali 480 Sep 12 02:25 contraband.zip
-rw-r--r-- 1 kali kali 0 Sep 12 05:06 gumbo1_output.txt
-rw-r--r-- 1 kali kali 0 Sep 12 05:06 gumbo2_output.txt
-rw-r--r-- 1 kali kali 40960 Sep 12 05:14 photorec.se2
drwxr-xr-x 2 kali kali 4096 Sep 12 05:17 recup_dir.1
drwxr-xr-x 2 kali kali 4096 Sep 12 05:18 recup_dir.2
-rw-r--r-- 1 kali kali 3187907 Apr 26 2004 rhino.log
-rw-r--r-- 1 kali kali 65703 Sep 12 01:57 rhino1.jpg
-rw-r--r-- 1 kali kali 292604 Apr 28 2004 rhino2.log
-rw-r--r-- 1 kali kali 720 Sep 12 02:19 rhino3.jpg
-rw-r--r-- 1 kali kali 226094 Apr 28 2004 rhino3.log
drwxr-xr-x 3 kali kali 4096 Sep 12 05:36 steg

(kali@aivtic-students-kali)-[~/LABS]
$
```

rhino2.log

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http contains ".jpg"

No.	Time	Source	Destination	Protocol	Length	Info
34	2004-04-28 21:08:35.819504	137.30.120.37	137.30.123.234	HTTP	1033	HTTP/1.1 200
49	2004-04-28 21:08:38.073111	137.30.123.234	137.30.120.37	HTTP	488	GET /~gnome/

Frame 34: 1033 bytes on wire (8264 bits), 1033 bytes captured (8264 bits)

- Ethernet II, Src: Oracle_72:6f:0b (08:00:20:72:6f:0b), Dst: Apple_cc:57:92 (00:03:93:cc:57:92)
- Internet Protocol Version 4, Src: 137.30.120.37, Dst: 137.30.123.234
- Transmission Control Protocol, Src Port: 80, Dst Port: 2026, Seq: 589, Ack: 768, Len: 979
- Hypertext Transfer Protocol, has 2 chunks (including last chunk)
- Line-based text data: text/html (15 lines)

rhino2.log Packets: 370 · Displayed: 2 (0.5%) Profile: Default

```
kali@aiutic-students-kali: ~/LABS
$ whois 137.30.123.234

#
# ARIN WHOIS data and services are subject to the Terms of Use
# available at: https://www.arin.net/resources/registry/whois/tou/
#
# If you see inaccuracies in the results, please report at
# https://www.arin.net/resources/registry/whois/inaccuracy_reporting/
#
# Copyright 1997-2026, American Registry for Internet Numbers, Ltd.
#

NetRange: 137.30.0.0 - 137.30.255.255
CIDR: 137.30.0.0/16
NetName: UNO-NET-BIG
NetHandle: NET-137-30-0-0-1
Parents: NET137 (NET-137-0-0-0-0)
NetType: Direct Allocation
OriginAS:
Organization: University of New Orleans (UNO)
RegDate: 1989-08-17
Updated: 2021-12-14
Ref: https://rdap.arin.net/registry/ip/137.30.0.0

OrgName: University of New Orleans
OrgId: UNO
Address: Computer Research Building
Address: Lakefront Campus
City: New Orleans
StateProv: LA
PostalCode: 70148
Country: US
RegDate: 1988-08-17
Updated: 2015-10-23
Ref: https://rdap.arin.net/registry/entity/UNO
```

```
File Edit View VM Tabs Help
Applications Places
Sep 12 07:16
kali@aiutic-students-kali: ~/LABS

OrgAbuseHandle: IEN2-ARIN
OrgAbuseName: IT Enterprise Networking
OrgAbusePhone: +1-504-288-1198
OrgAbuseEmail: hachoi@uno.edu
OrgAbuseRef: https://rdap.arin.net/registry/entity/IENT2-ARIN

OrgNOCHandle: IEN2-ARIN
OrgNOCHandle: IT Enterprise Networking
OrgNOCHandlePhone: +1-504-288-1198
OrgNOCHandleEmail: hachoi@uno.edu
OrgNOCHandleRef: https://rdap.arin.net/registry/entity/IENT2-ARIN

OrgTechHandle: IEN2-ARIN
OrgTechName: IT Enterprise Networking
OrgTechPhone: +1-504-288-1198
OrgTechEmail: hachoi@uno.edu
OrgTechRef: https://rdap.arin.net/registry/entity/IENT2-ARIN

RABuseHandle: IEN2-ARIN
RABuseName: IT Enterprise Networking
RABusePhone: +1-504-288-1198
RABuseEmail: hachoi@uno.edu
RABuseRef: https://rdap.arin.net/registry/entity/IENT2-ARIN

RNOCCHandle: IEN2-ARIN
RNOCCHandle: IT Enterprise Networking
RNOCCHandlePhone: +1-504-288-1198
RNOCCHandleEmail: hachoi@uno.edu
RNOCCHandleRef: https://rdap.arin.net/registry/entity/IENT2-ARIN

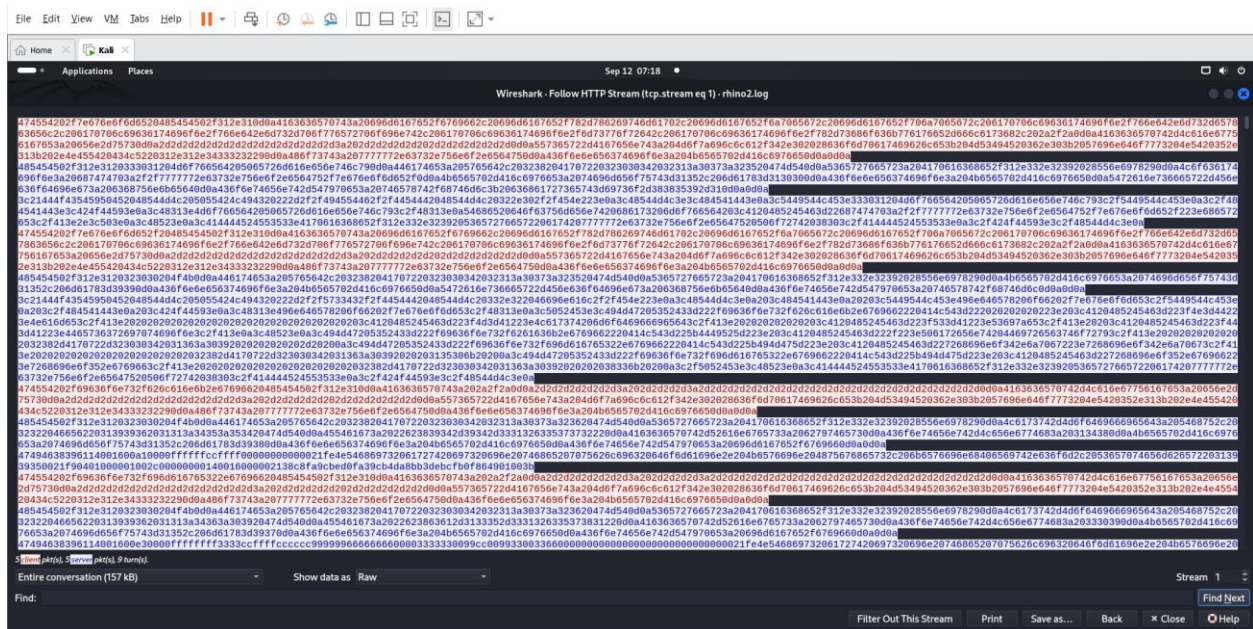
RTechHandle: IEN2-ARIN
RTechName: IT Enterprise Networking
RTechPhone: +1-504-288-1198
RTechEmail: hachoi@uno.edu
RTechRef: https://rdap.arin.net/registry/entity/IENT2-ARIN
```

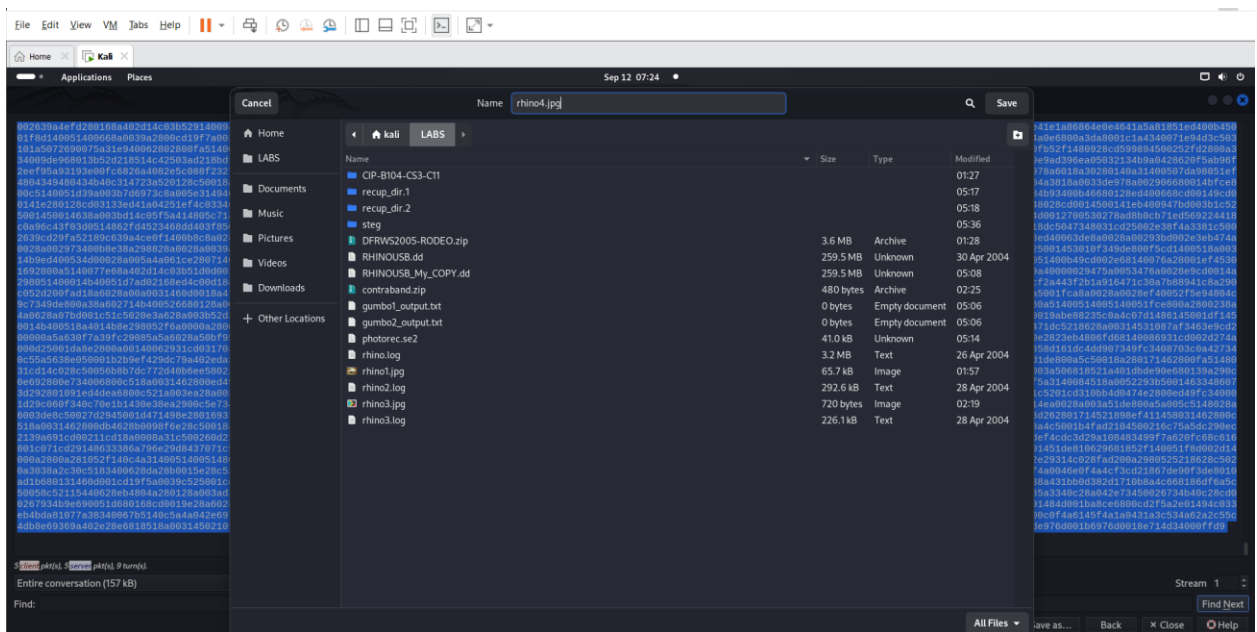
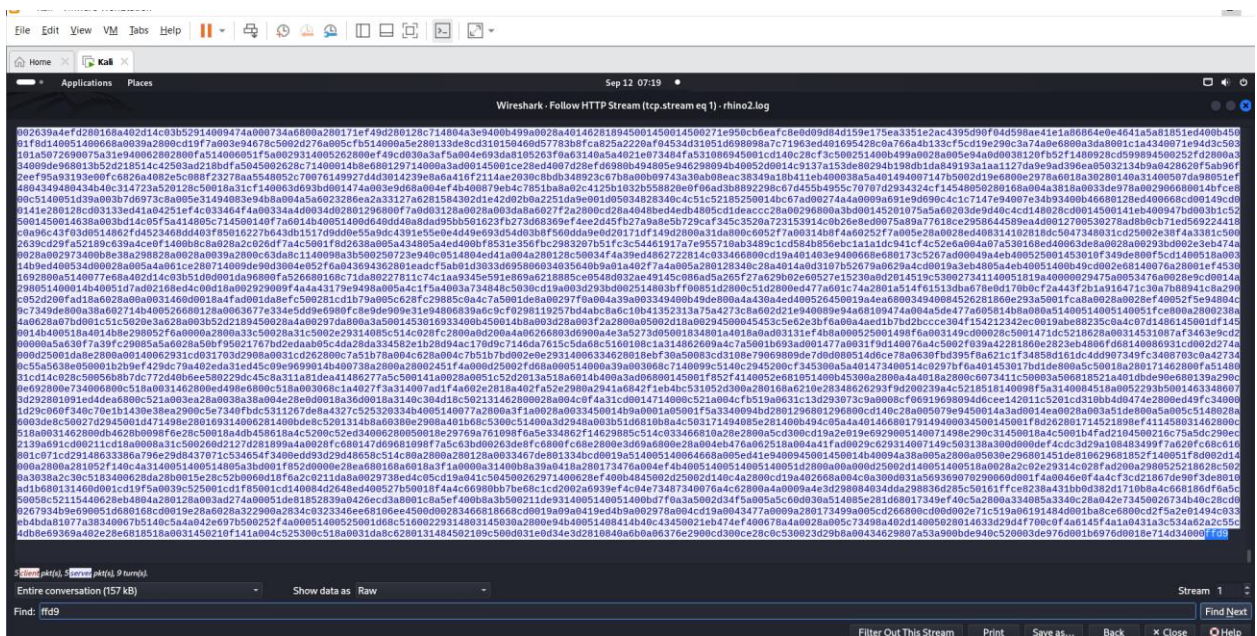
```
RABuseHandle: IEN2-ARIN
RABuseName: IT Enterprise Networking
RABusePhone: +1-504-288-1198
RABuseEmail: hachoi@uno.edu
RABuseRef: https://rdap.arin.net/registry/entity/IENT2-ARIN

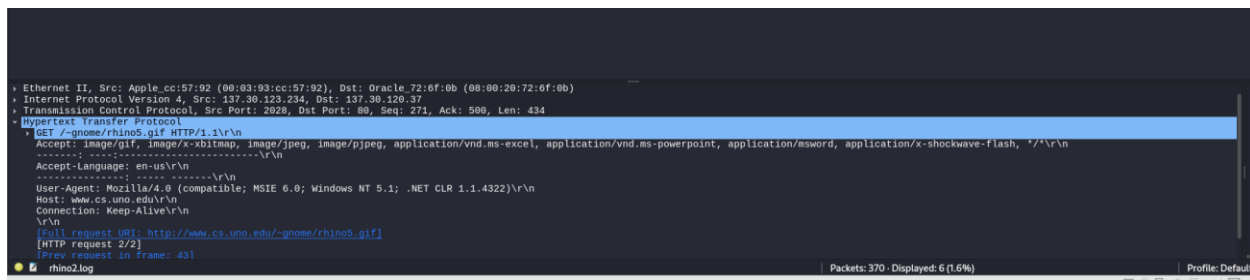
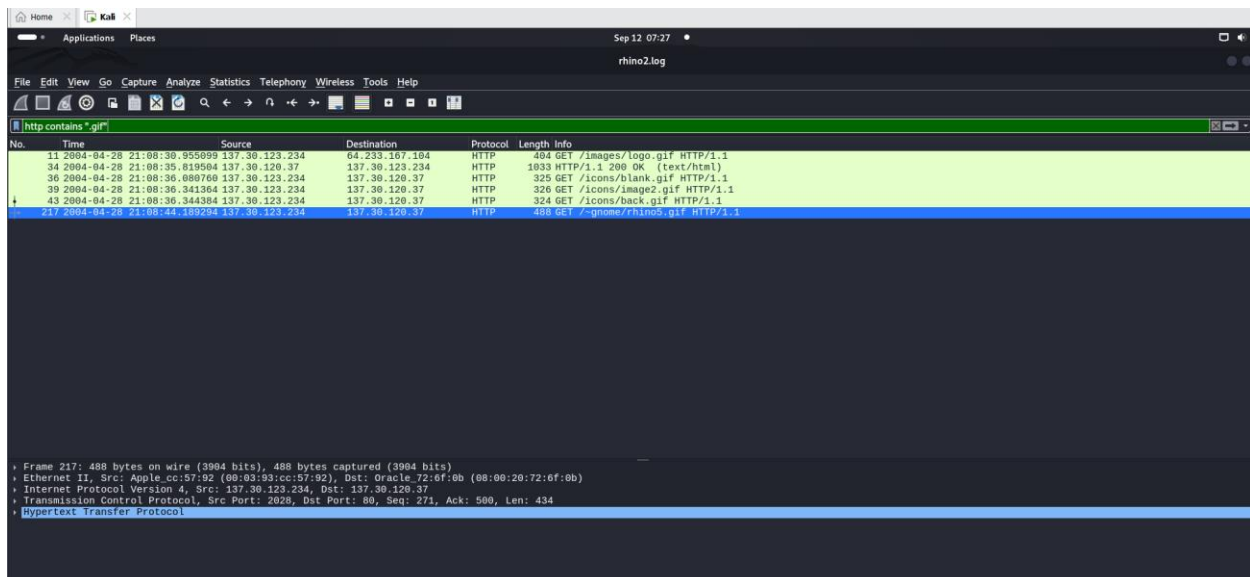
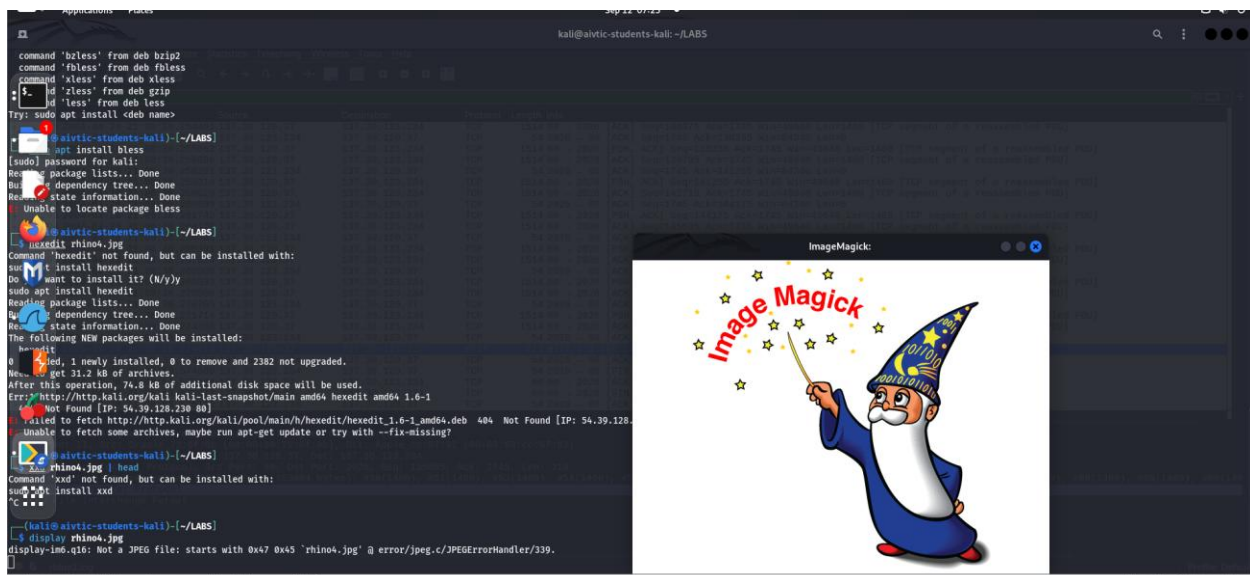
RNOCCHandle: IEN2-ARIN
RNOCCHandle: IT Enterprise Networking
RNOCCHandlePhone: +1-504-288-1198
RNOCCHandleEmail: hachoi@uno.edu
RNOCCHandleRef: https://rdap.arin.net/registry/entity/IENT2-ARIN

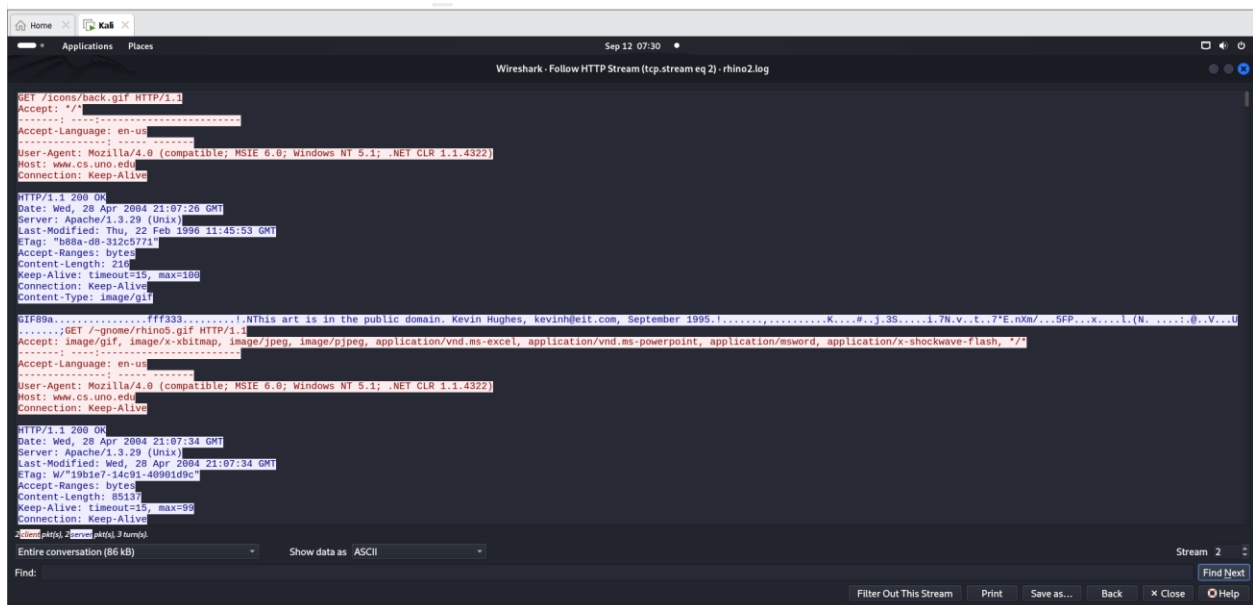
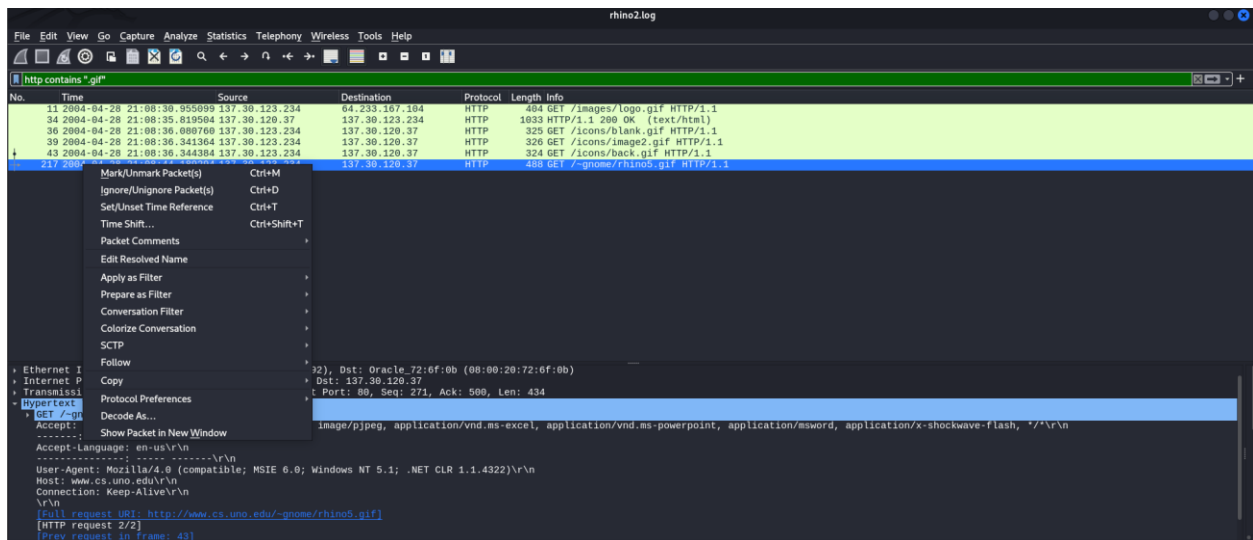
RTechHandle: IEN2-ARIN
RTechName: IT Enterprise Networking
RTechPhone: +1-504-288-1198
RTechEmail: hachoi@uno.edu
RTechRef: https://rdap.arin.net/registry/entity/IENT2-ARIN

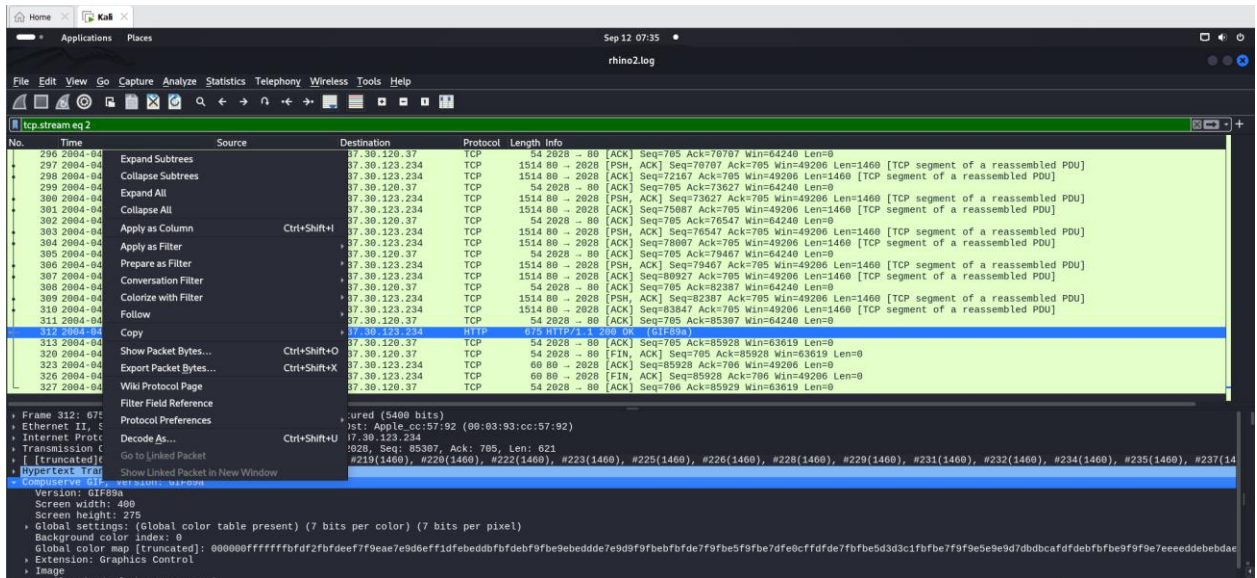
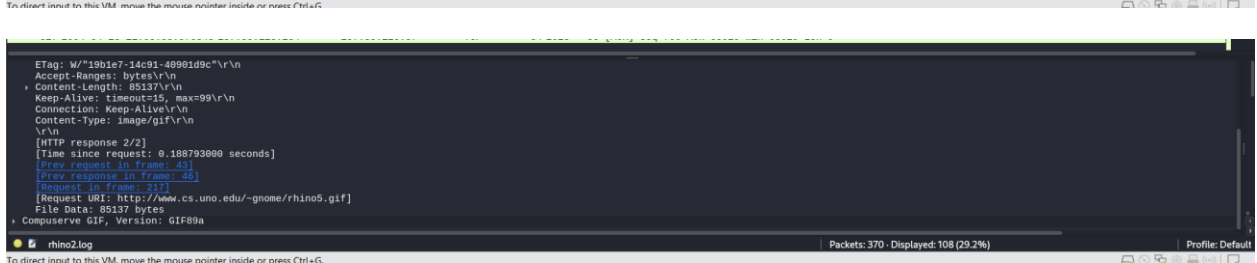
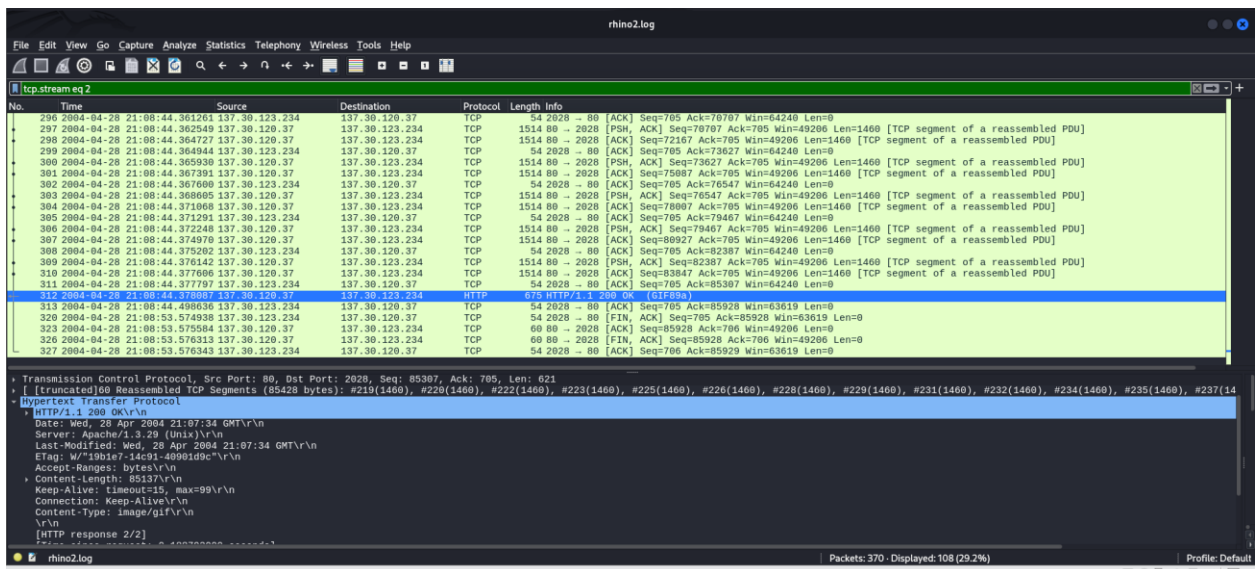
#
# ARIN WHOIS data and services are subject to the Terms of Use
# available at: https://www.arin.net/resources/registry/whois/tou/
#
# If you see inaccuracies in the results, please report at
# https://www.arin.net/resources/registry/whois/inaccuracy_reporting/
#
# Copyright 1997-2026, American Registry for Internet Numbers, Ltd.
#
```

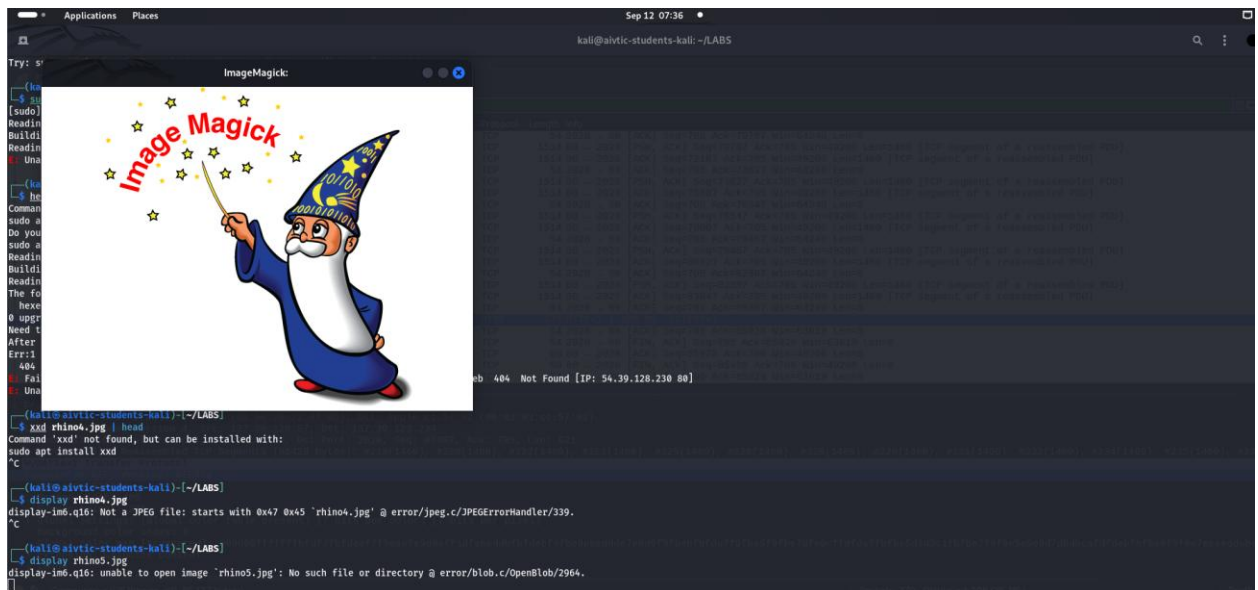
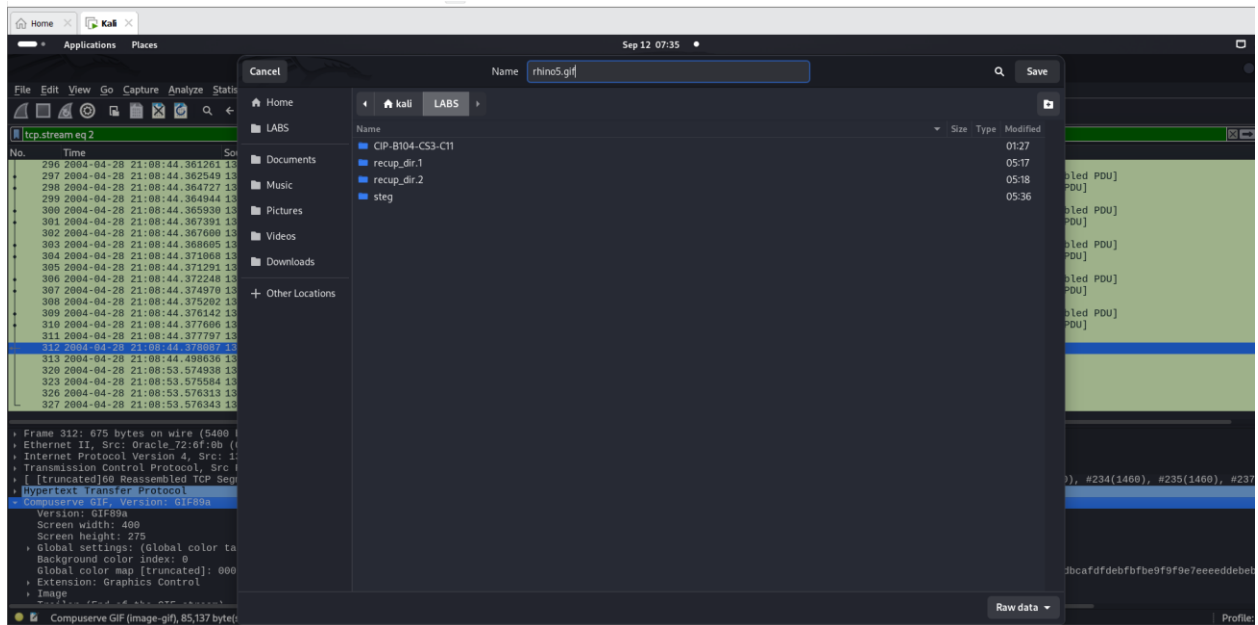












Go Home Applications Places Sep 12 07:38 rhino2.log

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http.content_type

No.	Time	Source	Destination	Protocol	Length	Info
30	2004-04-28 21:08:35.481949	137.30.120.37	137.30.123.234	HTTP	642	HTTP/1.1 301 Moved Permanently (text/html)
34	2004-04-28 21:08:35.819504	137.30.120.37	137.30.123.234	HTTP	1033	HTTP/1.1 200 OK (text/html)
37	2004-04-28 21:08:36.080585	137.30.120.37	137.30.123.234	HTTP	484	HTTP/1.1 200 OK (GIF89a)
45	2004-04-28 21:08:36.347255	137.30.120.37	137.30.123.234	HTTP	646	HTTP/1.1 200 OK (GIF89a)
46	2004-04-28 21:08:36.352379	137.30.120.37	137.30.123.234	HTTP	553	HTTP/1.1 200 OK (GIF89a)
215	2004-04-28 21:08:38.274614	137.30.120.37	137.30.123.234	HTTP	273	HTTP/1.1 200 OK (JPEG JFIF image)
312	2004-04-28 21:08:44.378087	137.30.120.37	137.30.123.234	HTTP	675	HTTP/1.1 200 OK (GIF89a)
345	2004-04-28 21:09:14.169172	137.30.120.37	137.30.123.234	HTTP	646	HTTP/1.1 301 Moved Permanently (text/html)
350	2004-04-28 21:09:14.458851	137.30.120.37	137.30.123.234	HTTP	267	HTTP/1.1 200 OK (text/html)
362	2004-04-28 21:09:57.137973	137.30.120.37	137.30.123.234	HTTP	1150	HTTP/1.1 200 OK (text/html)

Internet Protocol Version 4, Src: 137.30.120.37, Dst: 137.30.123.234

Transmission Control Protocol, Src Port: 80, Dst Port: 2020, Seq: 1560, Ack: 1039, Len: 430

Hypertext Transfer Protocol

HTTP/1.1 200 OK\r\n

Date: Wed, 28 Apr 2004 21:07:26 GMT\r\n

Server: Apache/1.3.29 (Unix)\r\n

Last-Modified: Thu, 22 Feb 1996 11:45:54 GMT\r\n

Etag: "b6ca-96-312e3772"\r\n

Accept-Ranges: bytes\r\n

Content-Length: 148\r\n

[Content length: 148]

Keep-Alive: timeout=15, max=98\r\n

Connection: Keep-Alive\r\n

Content-Type: image/gif\r\n

\r\n

[HTTP response 3/5]

Content-Type: Character string

Packets: 370 · Displayed: 10 (2.7%)

Profile: Default

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http.content_type

No.	Time	Source	Destination	Protocol	Length	Info
30	2004-04-28 21:08:35.481949	137.30.120.37	137.30.123.234	HTTP	642	HTTP/1.1 301 Moved Permanently (text/html)
34	2004-04-28 21:08:35.819504	137.30.120.37	137.30.123.234	HTTP	1033	HTTP/1.1 200 OK (text/html)
37	2004-04-28 21:08:36.080585	137.30.120.37	137.30.123.234	HTTP	484	HTTP/1.1 200 OK (GIF89a)
45	2004-04-28 21:08:36.347255	137.30.120.37	137.30.123.234	HTTP	646	HTTP/1.1 200 OK (GIF89a)
46	2004-04-28 21:08:36.352379	137.30.120.37	137.30.123.234	HTTP	553	HTTP/1.1 200 OK (GIF89a)
215	2004-04-28 21:08:38.274614	137.30.120.37	137.30.123.234	HTTP	273	HTTP/1.1 200 OK (JPEG JFIF image)
312	2004-04-28 21:08:44.378087	137.30.120.37	137.30.123.234	HTTP	675	HTTP/1.1 200 OK (GIF89a)
345	2004-04-28 21:09:14.169172	137.30.120.37	137.30.123.234	HTTP	646	HTTP/1.1 301 Moved Permanently (text/html)
350	2004-04-28 21:09:14.458851	137.30.120.37	137.30.123.234	HTTP	267	HTTP/1.1 200 OK (text/html)
362	2004-04-28 21:09:57.137973	137.30.120.37	137.30.123.234	HTTP	1150	HTTP/1.1 200 OK (text/html)

Internet Protocol Version 4, Src: 137.30.120.37, Dst: 137.30.123.234

Transmission Control Protocol, Src Port: 80, Dst Port: 2026, Seq: 1998, Ack: 1311, Len: 592

Hypertext Transfer Protocol

HTTP/1.1 200 OK\r\n

Date: Wed, 28 Apr 2004 21:07:26 GMT\r\n

Server: Apache/1.3.29 (Unix)\r\n

Last-Modified: Thu, 22 Feb 1996 11:46:09 GMT\r\n

Etag: "b6ca-126-312e3772"\r\n

Accept-Ranges: bytes\r\n

Content-Length: 309\r\n

[Content length: 309]

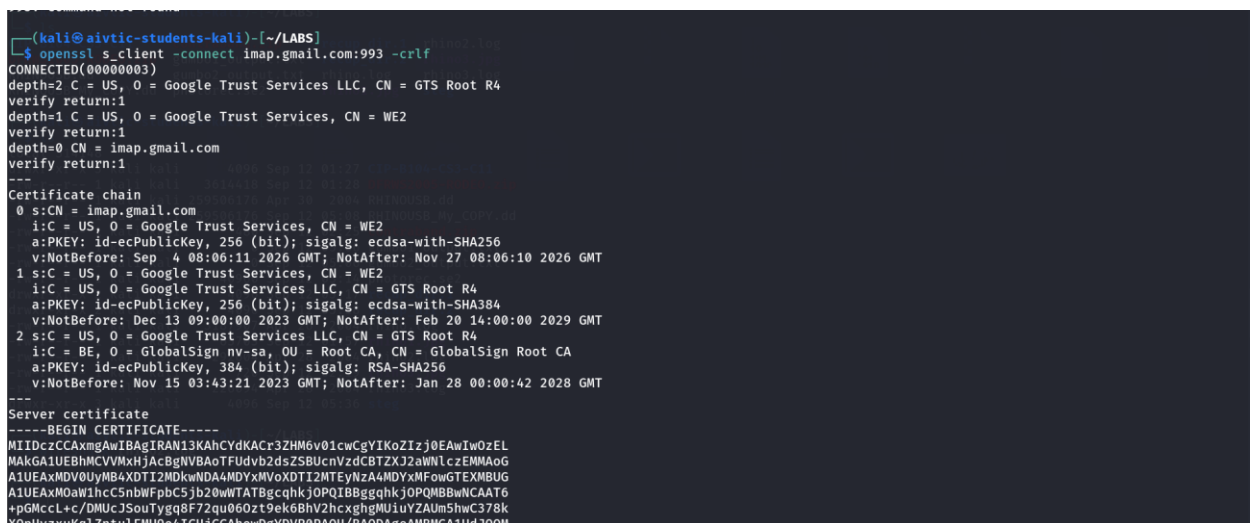
Keep-Alive: timeout=15, max=97\r\n

Connection: Keep-Alive\r\n

Content-Type: image/gif\r\n

\r\n

[HTTP response 4/5]




```
Applications Places Sep 12 08:40
kali@kali-students-kali: ~/LABS

Start Time: 1789216731
Timeout : 7200 (sec)
Verify return code: 0 (ok)
Extended master secret: no
Max Early Data: 0

---
read R BLOCK
---
Post-Handshake New Session Ticket arrived:
SSL-Session:
  Protocol : TLSv1.3
  Cipher : TLS_AES_256_GCM_SHA384
  Session-ID: FBCEA65E8F824D601DD351942F4F2E0E46921D1994ED88950790A56E0A44C53B
  Session-ID-ctx:
  Resumption PSK: D3494C37C6711D69C43AE8DC3B06472D928DD03311CF2E5119F5B3E2326629BD50609870A5967F1D711B9CCE4726E138
  PSK identity: None
  PSK identity hint: None
  SRP username: None
  TLS session ticket lifetime hint: 172800 (seconds)
  TLS session ticket:
0000 - 02 8c 51 fd 05 05 3a 67-c0 0a a2 10 e0 ed 93 ba ..Q...:g.....
0010 - cd 15 90 92 4f 1c 20 fc-3e ae 69 64 57 1a c2 30 ...0...>.idW..0
0020 - 7d dd 55 1f 21 83 54 92-51 60 09 2d 5a e6 47 2d }.U.!.T.Q'..-T.G-
0030 - 8f 46 d5 34 d0 ee 5e 8f-86 75 b4 9a 08 8e 36 20 .F.4...^..u...6
0040 - 79 58 98 dc 99 f2 f0 b3-bd 11 19 33 a3 ae dd 83 yX.....3....
0050 - 52 3e 62 35 a6 3d df 5c-07 b2 69 da 1a df e8 8b R>b5=.\.i.....
0060 - dc d9 e1 9a b4 d9 b0 28-18 83 ef 14 9d 9f 5a c6 .....(.....Z.
0070 - 23 5f 9c 69 7d a3 cb 03-41 36 6e 5e 34 c4 e9 a2 #..i}...A6n^4...
0080 - 39 15 0c 5b 1d 05 34 11-e6 39 5c bb f4 c1 87 f8 9..[...4..9\.....
0090 - a9 9a 11 9c fa 83 df 8d-81 16 ac b3 22 b7 ca f8 .....^.....'..
00a0 - 91 14 3a bc ac be 5c ea-93 dc d1 36 00 27 f5 b1 ...:...\....6...'
00b0 - e3 b5 4d 28 73 0d 41 f2-ef 74 77 35 e5 c8 19 90 ..M(s.A...tw5....
00c0 - 88 43 f4 d6 3c 06 1c f3-f3 18 72 f4 32 f3 61 75 .C.<.....r.2.au
```

```
Applications Places Sep 12 08:40
kali@kali-students-kali: ~/LABS

Session-ID: FBCEA65E8F824D601DD351942F4F2E0E46921D1994ED88950790A56E0A44C53B
Session-ID-ctx:
Resumption PSK: D3494C37C6711D69C43AE8DC3B06472D928DD03311CF2E5119F5B3E2326629BD50609870A5967F1D711B9CCE4726E138
PSK identity: None
PSK identity hint: None
SRP username: None
TLS session ticket lifetime hint: 172800 (seconds)
TLS session ticket:
0000 - 02 8c 51 fd 05 05 3a 67-c0 0a a2 10 e0 ed 93 ba ..Q...:g.....
0010 - cd 15 90 92 4f 1c 20 fc-3e ae 69 64 57 1a c2 30 ...0...>.idW..0
0020 - 7d dd 55 1f 21 83 54 92-51 60 09 2d 5a e6 47 2d }.U.!.T.Q'..-T.G-
0030 - 8f 46 d5 34 d0 ee 5e 8f-86 75 b4 9a 08 8e 36 20 .F.4...^..u...6
0040 - 79 58 98 dc 99 f2 f0 b3-bd 11 19 33 a3 ae dd 83 yX.....3....
0050 - 52 3e 62 35 a6 3d df 5c-07 b2 69 da 1a df e8 8b R>b5=.\.i.....
0060 - dc d9 e1 9a b4 d9 b0 28-18 83 ef 14 9d 9f 5a c6 .....(.....Z.
0070 - 23 5f 9c 69 7d a3 cb 03-41 36 6e 5e 34 c4 e9 a2 #..i}...A6n^4...
0080 - 39 15 0c 5b 1d 05 34 11-e6 39 5c bb f4 c1 87 f8 9..[...4..9\.....
0090 - a9 9a 11 9c fa 83 df 8d-81 16 ac b3 22 b7 ca f8 .....^.....'..
00a0 - 91 14 3a bc ac be 5c ea-93 dc d1 36 00 27 f5 b1 ...:...\....6...'
00b0 - e3 b5 4d 28 73 0d 41 f2-ef 74 77 35 e5 c8 19 90 ..M(s.A...tw5....
00c0 - 88 43 f4 d6 3c 06 1c f3-f3 18 72 f4 32 f3 61 75 .C.<.....r.2.au
00d0 - 69 e1 f3 2b 1b e3 a4 23-bb 44 3c 26 4b d2 90 35 i..+...#.D<6K..5
00e0 - de 10 73 ad 42 52 55 70-30 61 ca f3 2f b6 8d ..s.BRUp0a.../..

Start Time: 1789216731
Timeout : 7200 (sec)
Verify return code: 0 (ok)
Extended master secret: no
Max Early Data: 0

---
read R BLOCK
* OK Gimap ready for requests from 105.127.16.20 5b1f17b1804b1-49e64ca142bmb1044146735e9
```

```
---
Max Early Data: 0

---
read R BLOCK
* OK Gimap ready for requests from 105.127.16.20 5b1f17b1804b1-49e64ca142bmb1044146735e9
tag1 capability
* CAPABILITY IMAP4rev1 UNSELECT IDLE NAMESPACE QUOTA ID XLIST CHILDREN X-GM-EXT-1 XYZZY SASL-IR AUTH=XOAUTH2 AUTH=PLAIN AUTH=PLAIN-CLIENTTOKEN AUTH=OAUTHBEARE
R
tag1 OK Thats all she wrote! 5b1f17b1804b1-49e64ca142bmb1044146735e9
```

```
Start Time: 1789216731
Timeout : 7200 (sec)
Verify return code: 0 (ok)
Extended master secret: no
Max Early Data: 0

---
read R BLOCK
* OK Gimap ready for requests from 105.127.16.20 5b1f17b1804b1-49e64ca142bmb1044146735e9
tag1 capability
* CAPABILITY IMAP4rev1 UNSELECT IDLE NAMESPACE QUOTA ID XLIST CHILDREN X-GM-EXT-1 XYZZY SASL-IR AUTH=XOAUTH2 AUTH=PLAIN AUTH=PLAIN-CLIENTTOKEN AUTH=OAUTHBEARE
R
tag1 OK Thats all she wrote! 5b1f17b1804b1-49e64ca142bmb1044146735e9
tag2 LOGIN lab.bowie.edu@gmail.com
tag2 BAD Not enough arguments provided 5b1f17b1804b1-49e64ca142bmb1044146735e9
tag3 LIST "" "*"
tag3 BAD Unknown command 5b1f17b1804b1-49e64ca142bmb1044146735e9
```

Kali Linux terminal window showing a Wireshark capture of an IMAP4 stream. The interface includes a menu bar (File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, Help), a toolbar, and a status bar (Sep 12 08:56, rhino.log).

The main display area shows a list of captured packets (No., Time, Source, Destination, Protocol, Length, Info). The selected packet (No. 622) is an IMAP4 message from 137.30.120.39 to 137.30.122.253, containing a 76-byte request: 1 authenticate login.

The packet details pane shows the following structure:

- Frame 622: 76 bytes on wire (608 bits), 76 bytes captured (608 bits)
- Ethernet II, Src: Apple.cc:57:92 (08:00:58:0c:57:92), Dst: Oracle.co:fc:f2 (08:00:20:ce:fc:f2)
- Internet Protocol Version 4, Src: 137.30.122.253, Dst: 137.30.120.39
- Transmission Control Protocol, Src Port: 1652, Dst Port: 143, Seq: 1, Ack: 126, Len: 22
- Internet Message Access Protocol

Kali Linux terminal window showing a Wireshark capture of an IMAP4 stream. The interface includes a menu bar (File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, Help), a toolbar, and a status bar (Sep 12 08:57, rhino.log).

The main display area shows a list of captured packets (No., Time, Source, Destination, Protocol, Length, Info). The selected packet (No. 622) is an IMAP4 message from 137.30.120.39 to 137.30.122.253, containing a 76-byte request: 1 authenticate login.

The packet details pane shows the following structure:

- Frame 622: 76 bytes on wire (608 bits), 76 bytes captured (608 bits)
- Ethernet II, Src: Apple.cc:57:92 (08:00:58:0c:57:92), Dst: Oracle.co:fc:f2 (08:00:20:ce:fc:f2)
- Internet Protocol Version 4, Src: 137.30.122.253, Dst: 137.30.120.39
- Transmission Control Protocol, Src Port: 1652, Dst Port: 143, Seq: 1, Ack: 126, Len: 22
- Internet Message Access Protocol

No.	Time	Source	Destination	Protocol	Length	Info
1	2004-04-28 21:15:41.191431	137.30.123.234	64.233.167.99	TCP	62	2157 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
2	2004-04-28 21:15:41.259344	64.233.167.99	137.30.123.234	TCP	60	80 → 2157 [SYN, ACK] Seq=0 Ack=1 Win=1460 Len=0 MSS=1460
3	2004-04-28 21:15:41.259477	137.30.123.234	64.233.167.99	TCP	54	2157 → 80 [ACK] Seq=1 Ack=1 Win=64240 Len=0
4	2004-04-28 21:15:41.262422	137.30.123.234	64.233.167.99	HTTP	356	GET / HTTP/1.1
5	2004-04-28 21:15:41.368303	64.233.167.99	137.30.123.234	TCP	1484	80 → 2157 [ACK] Seq=1 Ack=303 Win=2920 Len=1430 [TCP segment of a reassembled PDU]
6	2004-04-28 21:15:41.368634	137.30.123.234	64.233.167.99	TCP	54	2157 → 80 [ACK] Seq=303 Ack=1431 Win=62810 Len=0
7	2004-04-28 21:15:41.368634	137.30.123.234	64.233.167.99	TCP	1484	[TCP Out-Of-Order] 80 → 2157 [ACK] Seq=1431 Ack=303 Win=2920 Len=1430
8	2004-04-28 21:15:41.369061	137.30.123.234	64.233.167.99	TCP	54	2157 → 80 [ACK] Seq=303 Ack=3076 Win=64240 Len=0
9	2004-04-28 21:15:48.234997	137.30.123.234	64.233.167.99	HTTP	592	GET /search?hl=en&ie=UTF-8&oe=UTF-8&q=rhino.exe HTTP/1.1
10	2004-04-28 21:15:48.338278	64.233.167.99	137.30.123.234	TCP	1484	80 → 2157 [ACK] Seq=3076 Ack=303 Win=2920 Len=1430 [TCP segment of a reassembled PDU]
11	2004-04-28 21:15:48.340208	64.233.167.99	137.30.123.234	TCP	1484	80 → 2157 [ACK] Seq=4506 Ack=303 Win=2920 Len=1430 [TCP segment of a reassembled PDU]
12	2004-04-28 21:15:48.340344	137.30.123.234	64.233.167.99	TCP	54	2157 → 80 [ACK] Seq=841 Ack=5936 Win=64240 Len=0
13	2004-04-28 21:15:48.340344	137.30.123.234	64.233.167.99	TCP	1464	80 → 2157 [PSH, ACK] Seq=5936 Ack=303 Win=2920 Len=1410 [TCP segment of a reassembled PDU]
14	2004-04-28 21:15:48.451624	64.233.167.99	137.30.123.234	TCP	54	2157 → 80 [ACK] Seq=841 Ack=7346 Win=62830 Len=0
15	2004-04-28 21:15:48.451882	137.30.123.234	64.233.167.99	TCP	1484	80 → 2157 [ACK] Seq=7346 Ack=303 Win=2920 Len=1430 [TCP segment of a reassembled PDU]
16	2004-04-28 21:15:48.574439	64.233.167.99	137.30.123.234	TCP	1484	80 → 2157 [ACK] Seq=8776 Ack=303 Win=2920 Len=1430 [TCP segment of a reassembled PDU]
17	2004-04-28 21:15:48.575630	64.233.167.99	137.30.123.234	TCP	54	2157 → 80 [ACK] Seq=841 Ack=10206 Win=64240 Len=0
18	2004-04-28 21:15:48.575813	137.30.123.234	64.233.167.99	TCP	1484	80 → 2157 [ACK] Seq=10206 Ack=303 Win=2920 Len=1430 [TCP segment of a reassembled PDU]
19	2004-04-28 21:15:48.575886	64.233.167.99	137.30.123.234	TCP	1484	80 → 2157 [ACK] Seq=10206 Ack=303 Win=2920 Len=1430 [TCP segment of a reassembled PDU]
20	2004-04-28 21:15:48.576951	137.30.123.234	64.233.167.99	TCP	54	2157 → 80 [ACK] Seq=841 Ack=11636 Win=62810 Len=0
21	2004-04-28 21:15:48.577332	64.233.167.99	137.30.123.234	TCP	1484	80 → 2157 [ACK] Seq=11636 Ack=303 Win=2920 Len=1430 [TCP segment of a reassembled PDU]
22	2004-04-28 21:15:48.577505	64.233.167.99	137.30.123.234	TCP	1484	80 → 2157 [ACK] Seq=13866 Ack=303 Win=2920 Len=1430 [TCP segment of a reassembled PDU]
23	2004-04-28 21:15:48.577679	137.30.123.234	64.233.167.99	TCP	54	2157 → 80 [ACK] Seq=841 Ack=14496 Win=64240 Len=0

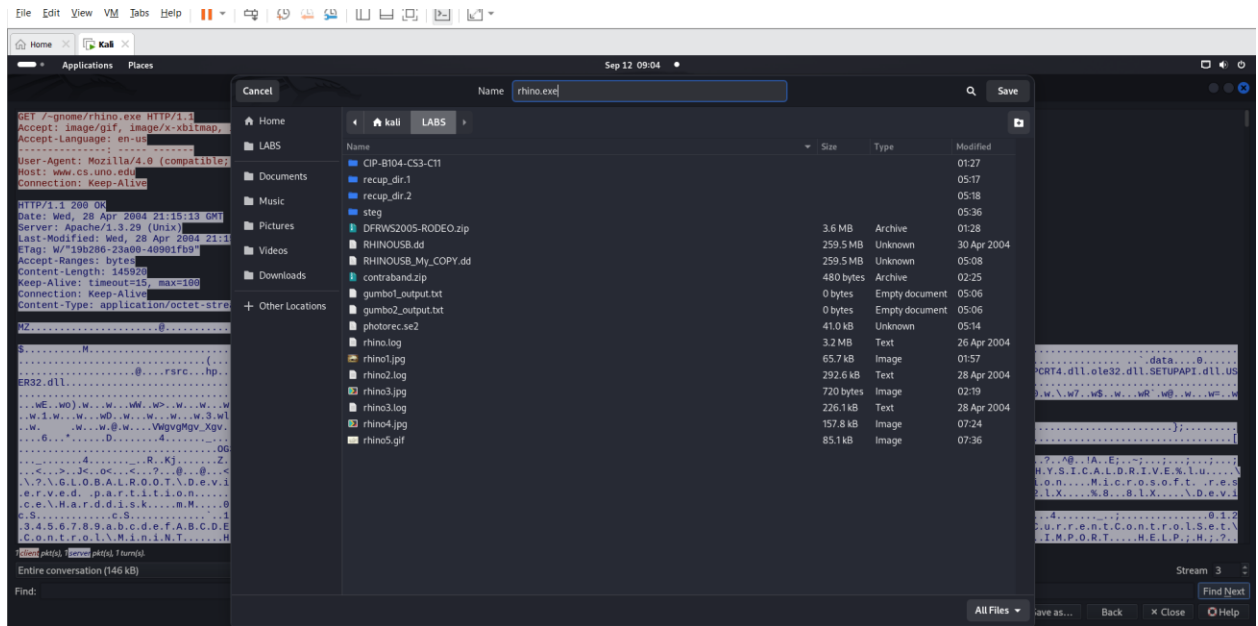
Frame 1: 62 bytes on wire (496 bits), 62 bytes captured (496 bits) on interface 0
 Ethernet II, Src: Apple.cc:17:02:00:03:93:cc:17:92, Dst: Cisco.41:a8:40:00:0d:ed:41:a0:40
 Internet Protocol Version 4, Src: 137.30.123.234, Dst: 64.233.167.99
 Transmission Control Protocol, Src Port: 2157, Dst Port: 80, Seq: 0, Len: 0

No.	Time	Source	Destination	Protocol	Length	Info
10	2004-04-28 21:15:48.234997	137.30.123.234	64.233.167.99	HTTP	592	GET /search?hl=en&ie=UTF-8&oe=UTF-8&q=rhino.exe HTTP/1.1
20	2004-04-28 21:15:48.325100	64.233.167.99	137.30.123.234	HTTP	1112	HTTP/1.1 200 OK (text/html)
59	2004-04-28 21:15:55.113910	137.30.123.234	216.239.51.99	HTTP	653	GET /groups?q=rhino.exe&hl=en&lr=de=UTF-8&oe=UTF-8&sa=N&tab=wg HTTP/1.1
85	2004-04-28 21:16:02.002505	137.30.123.234	216.239.51.99	HTTP	719	GET /groups?q=rhino.exe&hl=en&lr=de=UTF-8&oe=UTF-8&sel=MPG.159359f528ff3d2798aaca40news.ntworld.com&rnum=2 HTTP/1.1
110	2004-04-28 21:16:23.343061	137.30.123.234	137.30.120.37	HTTP	447	GET /-gnome/rhino.exe HTTP/1.1

Frame 10: 592 bytes on wire (4736 bits), 592 bytes captured (4736 bits) on interface 0
 Ethernet II, Src: Apple.cc:17:02:00:03:93:cc:17:92, Dst: Cisco.41:a8:40:00:0d:ed:41:a0:40
 Internet Protocol Version 4, Src: 137.30.123.234, Dst: 64.233.167.99
 Transmission Control Protocol, Src Port: 2157, Dst Port: 80, Seq: 303, Ack: 3076, Len: 536
 Hypertext Transfer Protocol

No.	Time	Source	Destination	Protocol	Length	Info
10	2004-04-28 21:15:48.234997	137.30.123.234	64.233.167.99	HTTP	592	GET /search?hl=en&ie=UTF-8&oe=UTF-8&q=rhino.exe HTTP/1.1
20	2004-04-28 21:15:48.325100	64.233.167.99	137.30.123.234	HTTP	1112	HTTP/1.1 200 OK (text/html)
59	2004-04-28 21:15:55.113910	137.30.123.234	216.239.51.99	HTTP	653	GET /groups?q=rhino.exe&hl=en&lr=de=UTF-8&oe=UTF-8&sa=N&tab=wg HTTP/1.1
85	2004-04-28 21:16:02.002505	137.30.123.234	216.239.51.99	HTTP	719	GET /groups?q=rhino.exe&hl=en&lr=de=UTF-8&oe=UTF-8&sel=MPG.159359f528ff3d2798aaca40news.ntworld.com&rnum=2 HTTP/1.1
110	2004-04-28 21:16:23.343061	137.30.123.234	137.30.120.37	HTTP	447	GET /-gnome/rhino.exe HTTP/1.1

Frame 110: 447 bytes on wire (3576 bits), 447 bytes captured (3576 bits) on interface 0
 Ethernet II, Src: Apple.cc:17:02:00:03:93:cc:17:92, Dst: Oracle.72:cf:0b:00:00:20:72:cf:0b
 Internet Protocol Version 4, Src: 137.30.123.234, Dst: 137.30.120.37
 Transmission Control Protocol, Src Port: 2163, Dst Port: 80, Seq: 1, Ack: 1, Len: 393
 Hypertext Transfer Protocol
 GET /-gnome/rhino.exe HTTP/1.1\r\n
 Accept: image/gif, image/x-bitmap, image/jpeg, image/pjpeg, application/vnd.ms-excel, application/vnd.ms-powerpoint, application/msword, application/x-shockwave-flash, */*\r\n
 Accept-Language: en-us\r\n
 -----\r\n
 User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; .NET CLR 1.1.4322)\r\n
 Host: www.cs.uno.edu\r\n
 Connection: Keep-Alive\r\n
 \r\n
 [Full request URI: http://www.cs.uno.edu/-gnome/rhino.exe]
 [HTTP request 1/2]
 [Response in frame: 274]



```
(kali@aivtic-students-kali)-[~/LABS]
$ ls -l rhino.exe
-rw-r--r-- 1 kali kali 146571 Sep 12 09:04 rhino.exe

(kali@aivtic-students-kali)-[~/LABS]
$ file rhino.exe
rhino.exe: ASCII text, with very long lines (8229)

(kali@aivtic-students-kali)-[~/LABS]
$ wine rhino.exe
Application could not be started, or no application associated with the specified file.
ShellExecuteEx failed: File not found.

(kali@aivtic-students-kali)-[~/LABS]
$ md5sum rhino.exe
2b95f7fc68a4a5ee5624e6f3492def66 rhino.exe

(kali@aivtic-students-kali)-[~/LABS]
```

PROFESSIONAL CONCLUSION

Based on the forensic examination conducted in Case ID C11/26/DFIT/17297, the available evidence demonstrates the presence and transmission of substantial rhinoceros-related digital material across multiple independent evidence sources. The investigation employed a multi-source forensic methodology involving file system analysis, FAT16 examination, file carving, network-traffic reconstruction, file-format validation, visual examination, cryptographic hashing, archive analysis, and cross-source correlation. Examination of RHINOUSB.dd established a FAT16 file system containing allocated textual files as well as recoverable image data. File-carving procedures recovered several valid image objects, including two images confirmed through visual examination to depict rhinoceroses. Other carved objects were identified as unrelated wildlife images, while duplicate objects were identified through identical cryptographic hashes and therefore were not treated as separate unique images. These findings demonstrate that examination of allocated file system entries alone would not have provided a complete representation of the digital content previously present on the USB device.

The network examination provided independent corroboration of rhinoceros-related digital activity. Analysis of rhino.log identified FTP sessions involving the transfer of rhino1.jpg, rhino3.jpg, and the encrypted contraband.zip archive. The recovered rhino1.jpg and rhino3.jpg files were validated as legitimate image files and visually confirmed to depict rhinoceroses. Two transfers associated with rhino3.jpg produced identical cryptographic hashes, establishing that they represented repeated transfers of the same digital object rather than two separate unique images. The encrypted contraband.zip archive was also recovered and structurally validated as a ZIP archive containing rhino2.jpg. Its encrypted status was confirmed, and its uncompressed

member size corresponded exactly with the size of one of the USB-carved JPEG objects. This represents a significant cross-source correlation, although definitive identification of the two objects as identical requires extraction of rhino2.jpg through the authorized recovery procedure followed by cryptographic hash comparison. Analysis of rhino2.log further identified HTTP requests and responses associated with rhino4.jpg and rhino5.gif. Both recovered objects were successfully reconstructed, validated, and visually confirmed as rhinoceros images, providing additional independent evidence of rhinoceros-related digital content within the examined network environment.

The investigation also recognizes the importance of evidentiary limitations, attribution, and proper interpretation. The presence of filenames, account identifiers, network sessions, search activity, or transferred files establishes technical activity within the examined evidence but does not, by itself, conclusively establish the identity, motive, ownership, or personal intent of a particular individual. Likewise, timestamps from file system metadata, archive records, and network captures represent different types of events and should be normalized and interpreted carefully within a UTC-based timeline. The steganographic component must similarly distinguish between a detector alert and confirmed hidden content; suspicious metadata, padding, statistical characteristics, or repeated strings constitute investigative leads unless hidden information is successfully extracted, validated, and correlated. The executable identified within rhino3.log should also be treated as network evidence rather than proof of execution, with any further examination limited to approve static-analysis procedures unless an appropriately isolated and authorized forensic environment is available.

Overall, the examination establishes a coherent body of corroborated rhinoceros-related digital evidence distributed across persistent storage and network communications. At the current stage, multiple distinct rhinoceros-image objects have been independently identified from the USB image, FTP traffic, and HTTP traffic, while duplicate occurrences have been excluded through cryptographic reduplications. The final determination of whether the case threshold of nine or more unique rhinoceros images has been satisfied should be based on the completed master provenance and reduplications table after the encrypted ZIP extraction and hash comparison, steganographic examination, executable documentation, and all remaining validation procedures have been completed. Subject to these final checks, the combination of recoverable image content, network-transfer evidence, file-format validation, cryptographic hashes, visual confirmation, and cross-source correlation provides a sound and technically defensible basis for concluding that multiple rhinoceros images were present within and/or transmitted through the examined digital environment. Any final opinion regarding personal possession or attribution should, however, remain appropriately qualified and clearly distinguish established forensic facts from investigative inference.